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Catheter system for introducing an expandable stent into the body of a patient

Abstract

This disclosure relates to a catheter system for introducing a heart valve stent into the body of a patient. The catheter system includes a catheter tip having a seat portion for accommodating the stent in its collapsed state and a stent holder for releasably fixing the stent, wherein the seat portion is constituted by a first sleeve and a second sleeve, said sleeves being moveable relative to each other and relative to the stent holder, and a catheter shaft for connecting the catheter tip to a handle.

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5755783	12/1997	Stobie et al.	N/A	N/A
5756476	12/1997	Epstein et al.	N/A	N/A
5758663	12/1997	Wilk et al.	N/A	N/A
5766151	12/1997	Valley et al.	N/A	N/A
5769780	12/1997	Hata et al.	N/A	N/A
5769812	12/1997	Stevens et al.	N/A	N/A
5769882	12/1997	Fogarty et al.	N/A	N/A
5769887	12/1997	Brown et al.	N/A	N/A
5772609	12/1997	Nguyen et al.	N/A	N/A
5776188	12/1997	Shepherd et al.	N/A	N/A
5782809	12/1997	Umeno et al.	N/A	N/A
5782904	12/1997	White et al.	N/A	N/A
5795331	12/1997	Cragg et al.	N/A	N/A
5797946	12/1997	Chin	N/A	N/A
5797960	12/1997	Stevens et al.	N/A	N/A
5799661	12/1997	Boyd et al.	N/A	N/A
5800456	12/1997	Maeda et al.	N/A	N/A
5800508	12/1997	Goicoechea et al.	N/A	N/A
5800531	12/1997	Cosgrove et al.	N/A	N/A
5807327	12/1997	Green et al.	N/A	N/A
5807384	12/1997	Mueller	N/A	N/A
5807405	12/1997	Vanney et al.	N/A	N/A
5810836	12/1997	Hussein et al.	N/A	N/A
5814016	12/1997	Valley et al.	N/A	N/A
5817113	12/1997	Gifford, III et al.	N/A	N/A
5817126	12/1997	Imran	N/A	N/A
5823956	12/1997	Roth et al.	N/A	N/A
5824037	12/1997	Fogarty et al.	N/A	N/A
5824038	12/1997	Wall	N/A	N/A
5824041	12/1997	Lenker et al.	N/A	N/A
5824043	12/1997	Cottone, Jr.	N/A	N/A
5824053	12/1997	Khosravi et al.	N/A	N/A
5824055	12/1997	Spiridigliozzi et al.	N/A	N/A
5824056	12/1997	Rosenberg	N/A	N/A
5824061	12/1997	Quijano et al.	N/A	N/A
5824063	12/1997	Cox	N/A	N/A
5824064	12/1997	Taheri	N/A	N/A
5824071	12/1997	Nelson et al.	N/A	N/A
5824080	12/1997	Lamuraglia	N/A	N/A
5829447	12/1997	Stevens et al.	N/A	N/A
5830222	12/1997	Makower	N/A	N/A
5836869	12/1997	Kudo et al.	N/A	N/A

5840081	12/1997	Andersen et al.	N/A	N/A
5841382	12/1997	Walden et al.	N/A	N/A
5843158	12/1997	Lenker et al.	N/A	N/A
5843161	12/1997	Solovay	N/A	N/A
5843181	12/1997	Jaffe et al.	N/A	N/A
5851232	12/1997	Lois	N/A	N/A
5853419	12/1997	Imran	N/A	N/A
5853420	12/1997	Chevillon et al.	N/A	N/A
5855210	12/1998	Sterman et al.	N/A	N/A
5855597	12/1998	Jayaraman	N/A	N/A
5855600	12/1998	Alt	N/A	N/A
5855601	12/1998	Bessler et al.	N/A	N/A
5855602	12/1998	Angell	N/A	N/A
5860966	12/1998	Tower	N/A	N/A
5860996	12/1998	Urban et al.	N/A	N/A
5861024	12/1998	Rashidi	N/A	N/A
5861028	12/1998	Angell	N/A	N/A
5865723	12/1998	Love	N/A	N/A
5868783	12/1998	Tower	N/A	N/A
5873812	12/1998	Ciana et al.	N/A	N/A
5873906	12/1998	Lau et al.	N/A	N/A
5876373	12/1998	Giba et al.	N/A	N/A
5876419	12/1998	Carpenter et al.	N/A	N/A
5876434	12/1998	Flomenblit et al.	N/A	N/A
5876448	12/1998	Thompson et al.	N/A	N/A
5878751	12/1998	Hussein et al.	N/A	N/A
5880242	12/1998	Hu et al.	N/A	N/A
5885228	12/1998	Rosenman et al.	N/A	N/A
5885238	12/1998	Stevens et al.	N/A	N/A
5885259	12/1998	Berg	N/A	N/A
5888201	12/1998	Stinson et al.	N/A	N/A
5891160	12/1998	Williamson, IV et al.	N/A	N/A
5891191	12/1998	Stinson	N/A	N/A
5895399	12/1998	Barbut et al.	N/A	N/A
5895420	12/1998	Mirsch, II et al.	N/A	N/A
5899936	12/1998	Goldstein	N/A	N/A
5906619	12/1998	Olson et al.	N/A	N/A
5907893	12/1998	Zadno-Azizi et al.	N/A	N/A
5908028	12/1998	Wilk	N/A	N/A
5908029	12/1998	Knudson et al.	N/A	N/A
5908451	12/1998	Yeo	N/A	N/A
5908452	12/1998	Bokros et al.	N/A	N/A
5910144	12/1998	Hayashi	N/A	N/A
5910154	12/1998	Tsugita et al.	N/A	N/A
5911734	12/1998	Tsugita et al.	N/A	N/A
5911752	12/1998	Dustrude et al.	N/A	N/A
5913842	12/1998	Boyd et al.	N/A	N/A
5916193	12/1998	Stevens et al.	N/A	N/A
5922022	12/1998	Nash et al.	N/A	N/A
5924424	12/1998	Stevens et al.	N/A	N/A

5925012	12/1998	Murphy-Chutorian et al.	N/A	N/A
5925063	12/1998	Khosravi	N/A	N/A
5928281	12/1998	Huynh et al.	N/A	N/A
5931848	12/1998	Saadat	N/A	N/A
5935119	12/1998	Guy et al.	N/A	N/A
5935161	12/1998	Robinson et al.	N/A	N/A
5935163	12/1998	Gabbay	N/A	N/A
5938632	12/1998	Ellis	N/A	N/A
5938697	12/1998	Killion et al.	N/A	N/A
5941908	12/1998	Goldsteen et al.	N/A	N/A
5944019	12/1998	Knudson et al.	N/A	N/A
5944738	12/1998	Amplatz et al.	N/A	N/A
5948017	12/1998	Taheri	N/A	N/A
5954764	12/1998	Parodi	N/A	N/A
5954766	12/1998	Zadno-Azizi et al.	N/A	N/A
5957949	12/1998	Leonhardt et al.	N/A	N/A
5961549	12/1998	Nguyen et al.	N/A	N/A
5964405	12/1998	Benary et al.	N/A	N/A
5964798	12/1998	Imran	N/A	N/A
5968064	12/1998	Selmon et al.	N/A	N/A
5968068	12/1998	Dehdashtian et al.	N/A	N/A
5968070	12/1998	Bley et al.	N/A	N/A
5971993	12/1998	Hussein et al.	N/A	N/A
5975949	12/1998	Holliday et al.	N/A	N/A
5976153	12/1998	Fischell et al.	N/A	N/A
5976155	12/1998	Foreman et al.	N/A	N/A
5976174	12/1998	Ruiz	N/A	N/A
5976178	12/1998	Goldsteen et al.	N/A	N/A
5976192	12/1998	McIntyre et al.	N/A	N/A
5976650	12/1998	Campbell et al.	N/A	N/A
5979455	12/1998	Maginot	N/A	N/A
5980455	12/1998	Daniel et al.	N/A	N/A
5980533	12/1998	Holman	N/A	N/A
5980548	12/1998	Evans et al.	N/A	N/A
5984956	12/1998	Tweden et al.	N/A	N/A
5984957	12/1998	Laptewicz, Jr. et al.	N/A	N/A
5984959	12/1998	Robertson et al.	N/A	N/A
5984964	12/1998	Roberts et al.	N/A	N/A
5987344	12/1998	West	N/A	N/A
5989276	12/1998	Houser et al.	N/A	N/A
5989287	12/1998	Yang et al.	N/A	N/A
5993469	12/1998	McKenzie et al.	N/A	N/A
5993481	12/1998	Marcade et al.	N/A	N/A
5997525	12/1998	March et al.	N/A	N/A
5997557	12/1998	Barbut et al.	N/A	N/A
5997563	12/1998	Kretzers	N/A	N/A
5997573	12/1998	Quijano et al.	N/A	N/A
5999678	12/1998	Murphy-Chutorian et al.	N/A	N/A

6001123	12/1998	Lau	N/A	N/A
6001126	12/1998	Nguyen-Thien-Nhon	N/A	N/A
6004261	12/1998	Sinofsky et al.	N/A	N/A
6004347	12/1998	McNamara et al.	N/A	N/A
6004348	12/1998	Banas et al.	N/A	N/A
6007543	12/1998	Ellis et al.	N/A	N/A
6010449	12/1999	Selmon et al.	N/A	N/A
6010522	12/1999	Barbut et al.	N/A	N/A
6010530	12/1999	Goicoechea	N/A	N/A
6010531	12/1999	Donlon et al.	N/A	N/A
6012457	12/1999	Lesh	N/A	N/A
6013854	12/1999	Moriuchi	N/A	N/A
6015431	12/1999	Thornton et al.	N/A	N/A
6019777	12/1999	MacKenzie	N/A	N/A
6019778	12/1999	Wilson et al.	N/A	N/A
6022370	12/1999	Tower	N/A	N/A
6026814	12/1999	LaFontaine et al.	N/A	N/A
6027476	12/1999	Sterman et al.	N/A	N/A
6027520	12/1999	Tsugita et al.	N/A	N/A
6027525	12/1999	Suh et al.	N/A	N/A
6029671	12/1999	Stevens et al.	N/A	N/A
6029672	12/1999	Vanney et al.	N/A	N/A
6033582	12/1999	Lee et al.	N/A	N/A
6035856	12/1999	LaFontaine et al.	N/A	N/A
6036677	12/1999	Javier, Jr. et al.	N/A	N/A
6036697	12/1999	DiCaprio	N/A	N/A
6042554	12/1999	Rosenman et al.	N/A	N/A
6042581	12/1999	Ryan et al.	N/A	N/A
6042589	12/1999	Marianne	N/A	N/A
6042598	12/1999	Tsugita et al.	N/A	N/A
6042607	12/1999	Williamson, IV et al.	N/A	N/A
6045565	12/1999	Ellis et al.	N/A	N/A
6051014	12/1999	Jang	N/A	N/A
6051104	12/1999	Oriaran et al.	N/A	N/A
6053924	12/1999	Hussein	N/A	N/A
6053942	12/1999	Eno et al.	N/A	N/A
6056743	12/1999	Ellis et al.	N/A	N/A
6059809	12/1999	Amor et al.	N/A	N/A
6059827	12/1999	Fenton, Jr.	N/A	N/A
6066160	12/1999	Colvin et al.	N/A	N/A
6067988	12/1999	Mueller	N/A	N/A
6068638	12/1999	Makower	N/A	N/A
6071292	12/1999	Makower et al.	N/A	N/A
6074416	12/1999	Berg et al.	N/A	N/A
6074417	12/1999	Peredo	N/A	N/A
6074418	12/1999	Buchanan et al.	N/A	N/A
6076529	12/1999	Vanney et al.	N/A	N/A
6076742	12/1999	Benary	N/A	N/A
6077297	12/1999	Robinson	623/1.11	A61F 2/95

6079414	12/1999	Roth	N/A	N/A
6080163	12/1999	Hussein et al.	N/A	N/A
6080170	12/1999	Nash et al.	N/A	N/A
6083257	12/1999	Taylor et al.	N/A	N/A
6091042	12/1999	Benary	N/A	N/A
6092526	12/1999	LaFontaine et al.	N/A	N/A
6092529	12/1999	Cox	N/A	N/A
6093166	12/1999	Knudson et al.	N/A	N/A
6093177	12/1999	Javier, Jr. et al.	N/A	N/A
6093185	12/1999	Ellis et al.	N/A	N/A
6093203	12/1999	Uflacker	N/A	N/A
6093530	12/1999	McIlroy et al.	N/A	N/A
6096074	12/1999	Pedros	N/A	N/A
6102941	12/1999	Tweden et al.	N/A	N/A
6102944	12/1999	Huynh et al.	N/A	N/A
6106550	12/1999	Magovern et al.	N/A	N/A
6110191	12/1999	Dehdashtian et al.	N/A	N/A
6110198	12/1999	Fogarty et al.	N/A	N/A
6110201	12/1999	Quijano et al.	N/A	N/A
6113612	12/1999	Swanson et al.	N/A	N/A
6113630	12/1999	Vanney et al.	N/A	N/A
6113823	12/1999	Eno	N/A	N/A
6117169	12/1999	Moe	N/A	N/A
6120520	12/1999	Saadat et al.	N/A	N/A
6120534	12/1999	Ruiz	N/A	N/A
6123682	12/1999	Knudson et al.	N/A	N/A
6123723	12/1999	Konya et al.	N/A	N/A
6125852	12/1999	Stevens et al.	N/A	N/A
6126649	12/1999	Vantassel et al.	N/A	N/A
6126654	12/1999	Giba et al.	N/A	N/A
6126685	12/1999	Lenker et al.	N/A	N/A
6126686	12/1999	Badylak et al.	N/A	N/A
6132451	12/1999	Payne et al.	N/A	N/A
6132473	12/1999	Williams et al.	N/A	N/A
6132986	12/1999	Pathak et al.	N/A	N/A
6139510	12/1999	Palermo	N/A	N/A
6139541	12/1999	Vanney et al.	N/A	N/A
6142987	12/1999	Tsugita	N/A	N/A
6143021	12/1999	Staehle	N/A	N/A
6143987	12/1999	Makita	N/A	N/A
6146366	12/1999	Schachar	N/A	N/A
6146415	12/1999	Fitz	N/A	N/A
6146417	12/1999	Ischinger	N/A	N/A
6152937	12/1999	Peterson et al.	N/A	N/A
6152956	12/1999	Pierce	N/A	N/A
6155264	12/1999	Ressemann et al.	N/A	N/A
6156031	12/1999	Aita et al.	N/A	N/A
6156055	12/1999	Ravenscroft	N/A	N/A
6156531	12/1999	Pathak et al.	N/A	N/A
6157852	12/1999	Selmon et al.	N/A	N/A

6159225	12/1999	Makower	N/A	N/A
6159239	12/1999	Greenhalgh	N/A	N/A
6162208	12/1999	Hipps	N/A	N/A
6162245	12/1999	Jayaraman	N/A	N/A
6165185	12/1999	Shennib et al.	N/A	N/A
6165188	12/1999	Saadat et al.	N/A	N/A
6165200	12/1999	Tsugita et al.	N/A	N/A
6165209	12/1999	Patterson et al.	N/A	N/A
6167605	12/2000	Morales	N/A	N/A
6168579	12/2000	Tsugita	N/A	N/A
6168614	12/2000	Andersen et al.	N/A	N/A
6168616	12/2000	Brown, III	N/A	N/A
6171251	12/2000	Mueller et al.	N/A	N/A
6171327	12/2000	Daniel et al.	N/A	N/A
6171335	12/2000	Wheatley et al.	N/A	N/A
6177514	12/2000	Pathak et al.	N/A	N/A
6179859	12/2000	Bates et al.	N/A	N/A
6182664	12/2000	Cosgrove	N/A	N/A
6182668	12/2000	Tweden et al.	N/A	N/A
6183481	12/2000	Lee et al.	N/A	N/A
6186972	12/2000	Nelson et al.	N/A	N/A
6187016	12/2000	Hedges et al.	N/A	N/A
6190353	12/2000	Makower et al.	N/A	N/A
6190393	12/2000	Bevier et al.	N/A	N/A
6190405	12/2000	Culombo et al.	N/A	N/A
6193726	12/2000	Vanney	N/A	N/A
6193734	12/2000	Bolduc et al.	N/A	N/A
6196230	12/2000	Hall et al.	N/A	N/A
6197050	12/2000	Eno et al.	N/A	N/A
6197053	12/2000	Cosgrove et al.	N/A	N/A
6197296	12/2000	Davies et al.	N/A	N/A
6197324	12/2000	Crittenden	N/A	N/A
6200311	12/2000	Danek et al.	N/A	N/A
6200336	12/2000	Pavcnik et al.	N/A	N/A
6203550	12/2000	Olson	N/A	N/A
6203556	12/2000	Evans et al.	N/A	N/A
6206888	12/2000	Bicek et al.	N/A	N/A
6206911	12/2000	Milo	N/A	N/A
6210408	12/2000	Chandrasekaran et al.	N/A	N/A
6210957	12/2000	Carpentier et al.	N/A	N/A
6213126	12/2000	LaFontaine et al.	N/A	N/A
6214036	12/2000	Letendre et al.	N/A	N/A
6214041	12/2000	Tweden et al.	N/A	N/A
6214054	12/2000	Cunanan et al.	N/A	N/A
6214055	12/2000	Simionescu et al.	N/A	N/A
6217527	12/2000	Selmon et al.	N/A	N/A
6217549	12/2000	Selmon et al.	N/A	N/A
6217575	12/2000	DeVore et al.	N/A	N/A
6217609	12/2000	Haverkost	N/A	N/A
6218662	12/2000	Tchakarov et al.	N/A	N/A

6221006	12/2000	Dubrul et al.	N/A	N/A
6221049	12/2000	Selmon et al.	N/A	N/A
6221091	12/2000	Khosravi	N/A	N/A
6221096	12/2000	Aiba et al.	N/A	N/A
6221100	12/2000	Strecker	N/A	N/A
6223752	12/2000	Vanney et al.	N/A	N/A
6224584	12/2000	March et al.	N/A	N/A
6231544	12/2000	Tsugita et al.	N/A	N/A
6231546	12/2000	Milo et al.	N/A	N/A
6231551	12/2000	Barbut	N/A	N/A
6231587	12/2000	Makower	N/A	N/A
6231602	12/2000	Carpentier et al.	N/A	N/A
6235000	12/2000	Milo et al.	N/A	N/A
6237607	12/2000	Vanney et al.	N/A	N/A
6238406	12/2000	Ellis et al.	N/A	N/A
6241667	12/2000	Vetter et al.	N/A	N/A
6241738	12/2000	Dereume	N/A	N/A
6241741	12/2000	Duhaylongsod et al.	N/A	N/A
6241757	12/2000	An et al.	N/A	N/A
6245102	12/2000	Jayaraman	N/A	N/A
6245103	12/2000	Stinson	N/A	N/A
6245105	12/2000	Nguyen et al.	N/A	N/A
6248112	12/2000	Gambale et al.	N/A	N/A
6248116	12/2000	Chevillon et al.	N/A	N/A
6250305	12/2000	Tweden	N/A	N/A
6251079	12/2000	Gambale et al.	N/A	N/A
6251104	12/2000	Kesten et al.	N/A	N/A
6251116	12/2000	Shennib et al.	N/A	N/A
6251135	12/2000	Stinson et al.	N/A	N/A
6251418	12/2000	Ahern et al.	N/A	N/A
6253768	12/2000	Wilk	N/A	N/A
6253769	12/2000	LaFONTAINE et al.	N/A	N/A
6254564	12/2000	Wilk et al.	N/A	N/A
6254635	12/2000	Schroeder et al.	N/A	N/A
6254636	12/2000	Peredo	N/A	N/A
6257634	12/2000	Wei	N/A	N/A
6258052	12/2000	Milo	N/A	N/A
6258087	12/2000	Edwards et al.	N/A	N/A
6258114	12/2000	Konya et al.	N/A	N/A
6258115	12/2000	Dubrul	N/A	N/A
6258119	12/2000	Hussein et al.	N/A	N/A
6258120	12/2000	McKenzie et al.	N/A	N/A
6258129	12/2000	Dybdal et al.	N/A	N/A
6258150	12/2000	MacKellar	N/A	N/A
6261304	12/2000	Hall et al.	N/A	N/A
6267783	12/2000	Letendre et al.	N/A	N/A
6269819	12/2000	Oz et al.	N/A	N/A
6270513	12/2000	Tsugita et al.	N/A	N/A
6270521	12/2000	Fischell et al.	N/A	N/A
6270526	12/2000	Cox	N/A	N/A

6273876	12/2000	Klima et al.	N/A	N/A
6273895	12/2000	Pinchuk et al.	N/A	N/A
6276661	12/2000	Laird	N/A	N/A
6277555	12/2000	Duran et al.	N/A	N/A
6283127	12/2000	Sterman et al.	N/A	N/A
6283951	12/2000	Flaherty et al.	N/A	N/A
6283983	12/2000	Makower et al.	N/A	N/A
6283995	12/2000	Moe et al.	N/A	N/A
6285903	12/2000	Rosenthal et al.	N/A	N/A
6287317	12/2000	Makower et al.	N/A	N/A
6287334	12/2000	Moll et al.	N/A	N/A
6287338	12/2000	Sarnowski et al.	N/A	N/A
6287339	12/2000	Vazquez et al.	N/A	N/A
6290709	12/2000	Ellis et al.	N/A	N/A
6290728	12/2000	Phelps et al.	N/A	N/A
6296662	12/2000	Caffey	N/A	N/A
6299637	12/2000	Shaolian et al.	N/A	N/A
6302875	12/2000	Makower et al.	N/A	N/A
6302892	12/2000	Wilk	N/A	N/A
6302906	12/2000	Goicoechea et al.	N/A	N/A
6306164	12/2000	Kujawski	N/A	N/A
6309382	12/2000	Garrison et al.	N/A	N/A
6309417	12/2000	Spence et al.	N/A	N/A
6311693	12/2000	Sterman et al.	N/A	N/A
6312465	12/2000	Griffin et al.	N/A	N/A
6319281	12/2000	Patel	N/A	N/A
6322548	12/2000	Payne et al.	N/A	N/A
6322593	12/2000	Pathak et al.	N/A	N/A
6325067	12/2000	Sterman et al.	N/A	N/A
6327772	12/2000	Zadno-Azizi et al.	N/A	N/A
6330884	12/2000	Kim	N/A	N/A
6331189	12/2000	Wolinsky et al.	N/A	N/A
6334873	12/2001	Lane et al.	N/A	N/A
6336934	12/2001	Gilson et al.	N/A	N/A
6336937	12/2001	Vonesh et al.	N/A	N/A
6338735	12/2001	Stevens	N/A	N/A
6338740	12/2001	Carpentier	N/A	N/A
6342070	12/2001	Nguyen-Thien-Nhon	N/A	N/A
6344044	12/2001	Fulkerson et al.	N/A	N/A
6346074	12/2001	Roth	N/A	N/A
6346116	12/2001	Brooks et al.	N/A	N/A
6348063	12/2001	Yassour et al.	N/A	N/A
6350248	12/2001	Knudson et al.	N/A	N/A
6350277	12/2001	Kocur	N/A	N/A
6350278	12/2001	Lenker et al.	N/A	N/A
6352547	12/2001	Brown et al.	N/A	N/A
6352554	12/2001	De Paulis	N/A	N/A
6352708	12/2001	Duran et al.	N/A	N/A
6357104	12/2001	Myers	N/A	N/A
6358277	12/2001	Duran	N/A	N/A

6361519	12/2001	Knudson et al.	N/A	N/A
6361545	12/2001	Macoviak et al.	N/A	N/A
6363938	12/2001	Saadat et al.	N/A	N/A
6363939	12/2001	Wilk	N/A	N/A
6364895	12/2001	Greenhalgh	N/A	N/A
6368338	12/2001	Konya et al.	N/A	N/A
6371970	12/2001	Khosravi et al.	N/A	N/A
6371983	12/2001	Lane	N/A	N/A
6375615	12/2001	Flaherty et al.	N/A	N/A
6378221	12/2001	Ekholm, Jr. et al.	N/A	N/A
6379319	12/2001	Garibotto et al.	N/A	N/A
6379365	12/2001	Diaz	N/A	N/A
6379372	12/2001	Dehdashtian et al.	N/A	N/A
6379383	12/2001	Palmaz et al.	N/A	N/A
6379740	12/2001	Rinaldi et al.	N/A	N/A
6380457	12/2001	Yurek et al.	N/A	N/A
6383193	12/2001	Cathcart et al.	N/A	N/A
6387119	12/2001	Wolf et al.	N/A	N/A
6387122	12/2001	Cragg	N/A	N/A
6390098	12/2001	LaFontaine et al.	N/A	N/A
6391051	12/2001	Sullivan, III et al.	N/A	N/A
6391538	12/2001	Vyavahare et al.	N/A	N/A
6395208	12/2001	Herweck et al.	N/A	N/A
6398807	12/2001	Chouinard et al.	N/A	N/A
6401720	12/2001	Stevens et al.	N/A	N/A
6402736	12/2001	Brown et al.	N/A	N/A
6402740	12/2001	Ellis et al.	N/A	N/A
6406488	12/2001	Tweden et al.	N/A	N/A
6406491	12/2001	Vanney	N/A	N/A
6406493	12/2001	Tu et al.	N/A	N/A
6409697	12/2001	Eno et al.	N/A	N/A
6409750	12/2001	Hyodoh et al.	N/A	N/A
6409751	12/2001	Hall et al.	N/A	N/A
6409755	12/2001	Vrba	N/A	N/A
6409759	12/2001	Peredo	N/A	N/A
6413275	12/2001	Nguyen et al.	N/A	N/A
6416490	12/2001	Ellis et al.	N/A	N/A
6416510	12/2001	Altman et al.	N/A	N/A
6423089	12/2001	Gingras et al.	N/A	N/A
6425916	12/2001	Garrison et al.	N/A	N/A
6432119	12/2001	Saadat	N/A	N/A
6432126	12/2001	Gambale et al.	N/A	N/A
6432127	12/2001	Kim et al.	N/A	N/A
6432132	12/2001	Cottone et al.	N/A	N/A
6440164	12/2001	DiMatteo et al.	N/A	N/A
6443158	12/2001	LaFontaine et al.	N/A	N/A
6447522	12/2001	Gambale et al.	N/A	N/A
6447539	12/2001	Nelson et al.	N/A	N/A
6451025	12/2001	Jervis	N/A	N/A
6451054	12/2001	Stevens	N/A	N/A

6454760	12/2001	Vanney	N/A	N/A
6454794	12/2001	Knudson et al.	N/A	N/A
6454799	12/2001	Schreck	N/A	N/A
6458092	12/2001	Gambale et al.	N/A	N/A
6458140	12/2001	Akin et al.	N/A	N/A
6458153	12/2001	Bailey et al.	N/A	N/A
6458323	12/2001	Boekstegers	N/A	N/A
6461382	12/2001	Cao	N/A	N/A
6464709	12/2001	Shennib et al.	N/A	N/A
6468303	12/2001	Amplatz et al.	N/A	N/A
6468660	12/2001	Ogle et al.	N/A	N/A
6471723	12/2001	Ashworth et al.	N/A	N/A
6475169	12/2001	Ferrera	N/A	N/A
6475226	12/2001	Belef et al.	N/A	N/A
6475239	12/2001	Campbell et al.	N/A	N/A
6475244	12/2001	Herweck et al.	N/A	N/A
6478819	12/2001	Moe	N/A	N/A
6479079	12/2001	Pathak et al.	N/A	N/A
6482220	12/2001	Mueller	N/A	N/A
6482228	12/2001	Norred	N/A	N/A
6485501	12/2001	Green	N/A	N/A
6485502	12/2001	Don Michael et al.	N/A	N/A
6485513	12/2001	Fan	N/A	N/A
6485524	12/2001	Strecker	N/A	N/A
6487581	12/2001	Spence et al.	N/A	N/A
6488704	12/2001	Connelly et al.	N/A	N/A
6491689	12/2001	Ellis et al.	N/A	N/A
6491707	12/2001	Makower et al.	N/A	N/A
6494211	12/2001	Boyd et al.	N/A	N/A
6494897	12/2001	Sterman et al.	N/A	N/A
6494909	12/2001	Greenhalgh	N/A	N/A
6503272	12/2002	Duerig et al.	N/A	N/A
6508496	12/2002	Huang	N/A	N/A
6508803	12/2002	Horikawa et al.	N/A	N/A
6508825	12/2002	Selmon et al.	N/A	N/A
6508833	12/2002	Pavcnik et al.	N/A	N/A
6509145	12/2002	Torrianni	N/A	N/A
6511458	12/2002	Milo et al.	N/A	N/A
6511491	12/2002	Grudem et al.	N/A	N/A
6514217	12/2002	Selmon et al.	N/A	N/A
6514271	12/2002	Evans et al.	N/A	N/A
6517527	12/2002	Gambale et al.	N/A	N/A
6517558	12/2002	Gittings et al.	N/A	N/A
6517573	12/2002	Pollock et al.	N/A	N/A
6521179	12/2002	Girardot et al.	N/A	N/A
6524323	12/2002	Nash et al.	N/A	N/A
6524335	12/2002	Hartley et al.	N/A	N/A
6527800	12/2002	McGuckin, Jr. et al.	N/A	N/A
6530949	12/2002	Konya et al.	N/A	N/A
6530952	12/2002	Vesely	N/A	N/A

6533807	12/2002	Wolinsky et al.	N/A	N/A
6537297	12/2002	Tsugita et al.	N/A	N/A
6537310	12/2002	Palmaz et al.	N/A	N/A
6540768	12/2002	Diaz et al.	N/A	N/A
6540782	12/2002	Snyders	N/A	N/A
6544230	12/2002	Flaherty et al.	N/A	N/A
6547827	12/2002	Carpentier et al.	N/A	N/A
6551303	12/2002	Van Tassel et al.	N/A	N/A
6558318	12/2002	Daniel et al.	N/A	N/A
6558417	12/2002	Peredo	N/A	N/A
6558418	12/2002	Carpentier et al.	N/A	N/A
6558429	12/2002	Taylor	N/A	N/A
6559132	12/2002	Holmer	N/A	N/A
6561998	12/2002	Roth et al.	N/A	N/A
6562031	12/2002	Chandrasekaran et al.	N/A	N/A
6562058	12/2002	Seguin et al.	N/A	N/A
6562063	12/2002	Euteneuer et al.	N/A	N/A
6562069	12/2002	Cai et al.	N/A	N/A
6564805	12/2002	Garrison et al.	N/A	N/A
6565528	12/2002	Mueller	N/A	N/A
6565594	12/2002	Herweck et al.	N/A	N/A
6569145	12/2002	Shmulewitz et al.	N/A	N/A
6569147	12/2002	Evans et al.	N/A	N/A
6569196	12/2002	Vesely	N/A	N/A
6572593	12/2002	Daum	N/A	N/A
6572642	12/2002	Rinaldi et al.	N/A	N/A
6572643	12/2002	Gharibadeh	N/A	N/A
6572652	12/2002	Shaknovich	N/A	N/A
6575168	12/2002	LaFontaine et al.	N/A	N/A
6579311	12/2002	Makower	N/A	N/A
6582444	12/2002	Wilk	N/A	N/A
6582460	12/2002	Cryer	N/A	N/A
6582462	12/2002	Andersen et al.	N/A	N/A
6585756	12/2002	Strecker	N/A	N/A
6585758	12/2002	Chouinard et al.	N/A	N/A
6585766	12/2002	Huynh et al.	N/A	N/A
6589279	12/2002	Anderson et al.	N/A	N/A
6592546	12/2002	Barbut et al.	N/A	N/A
6592614	12/2002	Lenker et al.	N/A	N/A
6599304	12/2002	Selmon et al.	N/A	N/A
6600803	12/2002	Bruder et al.	N/A	N/A
6605053	12/2002	Kamm et al.	N/A	N/A
6605112	12/2002	Moll et al.	N/A	N/A
6605113	12/2002	Wilk	N/A	N/A
6608040	12/2002	Lin et al.	N/A	N/A
6610077	12/2002	Hancock et al.	N/A	N/A
6610085	12/2002	Lazarus	N/A	N/A
6610100	12/2002	Phelps et al.	N/A	N/A
6613069	12/2002	Boyd et al.	N/A	N/A
6613077	12/2002	Gilligan et al.	N/A	N/A

6613079	12/2002	Wolinsky et al.	N/A	N/A
6613081	12/2002	Kim et al.	N/A	N/A
6613086	12/2002	Moe et al.	N/A	N/A
6616675	12/2002	Evard et al.	N/A	N/A
6616682	12/2002	Joergensen et al.	N/A	N/A
6622604	12/2002	Chouinard et al.	N/A	N/A
6623491	12/2002	Thompson	N/A	N/A
6623518	12/2002	Thompson et al.	N/A	N/A
6623521	12/2002	Steinke et al.	N/A	N/A
6626938	12/2002	Butaric et al.	N/A	N/A
6626939	12/2002	Burnside et al.	N/A	N/A
6632241	12/2002	Hancock et al.	N/A	N/A
6632243	12/2002	Zadno-Azizi et al.	N/A	N/A
6632470	12/2002	Morra et al.	N/A	N/A
6635068	12/2002	Dubrul et al.	N/A	N/A
6635079	12/2002	Unsworth et al.	N/A	N/A
6635080	12/2002	Lauterjung et al.	N/A	N/A
6635085	12/2002	Caffey et al.	N/A	N/A
6638237	12/2002	Guiles et al.	N/A	N/A
6638247	12/2002	Selmon et al.	N/A	N/A
6638293	12/2002	Makower et al.	N/A	N/A
6641610	12/2002	Wolf et al.	N/A	N/A
6651670	12/2002	Rapacki et al.	N/A	N/A
6651672	12/2002	Roth	N/A	N/A
6652540	12/2002	Cole et al.	N/A	N/A
6652546	12/2002	Nash et al.	N/A	N/A
6652555	12/2002	Vantassel et al.	N/A	N/A
6652571	12/2002	White et al.	N/A	N/A
6652578	12/2002	Bailey et al.	N/A	N/A
6655386	12/2002	Makower et al.	N/A	N/A
6656213	12/2002	Solem	N/A	N/A
6660003	12/2002	DeVore et al.	N/A	N/A
6660024	12/2002	Flaherty et al.	N/A	N/A
6663588	12/2002	DuBois et al.	N/A	N/A
6663663	12/2002	Kim et al.	N/A	N/A
6663667	12/2002	Dehdashtian et al.	N/A	N/A
6666863	12/2002	Wentzel et al.	N/A	N/A
6669709	12/2002	Cohn et al.	N/A	N/A
6669724	12/2002	Park et al.	N/A	N/A
6673089	12/2003	Yassour et al.	N/A	N/A
6673101	12/2003	Fitzgerald et al.	N/A	N/A
6673106	12/2003	Mitelberg et al.	N/A	N/A
6673109	12/2003	Cox	N/A	N/A
6676668	12/2003	Mercereau et al.	N/A	N/A
6676692	12/2003	Rabkin et al.	N/A	N/A
6676693	12/2003	Belding et al.	N/A	N/A
6676698	12/2003	McGuckin, Jr. et al.	N/A	N/A
6679268	12/2003	Stevens et al.	N/A	N/A
6682543	12/2003	Barbut et al.	N/A	N/A
6682558	12/2003	Tu et al.	N/A	N/A

6682559	12/2003	Myers et al.	N/A	N/A
6685648	12/2003	Flaherty et al.	N/A	N/A
6685739	12/2003	DiMatteo et al.	N/A	N/A
6689144	12/2003	Gerberding	N/A	N/A
6689164	12/2003	Seguin	N/A	N/A
6692512	12/2003	Jang	N/A	N/A
6692513	12/2003	Streeter et al.	N/A	N/A
6694983	12/2003	Wolf et al.	N/A	N/A
6695864	12/2003	Macoviak et al.	N/A	N/A
6695865	12/2003	Boyle et al.	N/A	N/A
6695875	12/2003	Stelter et al.	N/A	N/A
6695878	12/2003	McGuckin, Jr. et al.	N/A	N/A
6699274	12/2003	Stinson	N/A	N/A
6701932	12/2003	Knudson et al.	N/A	N/A
6702851	12/2003	Chinn et al.	N/A	N/A
6709425	12/2003	Gambale et al.	N/A	N/A
6709444	12/2003	Makower	N/A	N/A
6712842	12/2003	Gifford, III et al.	N/A	N/A
6712843	12/2003	Elliott	N/A	N/A
6714841	12/2003	Wright et al.	N/A	N/A
6714842	12/2003	Ito	N/A	N/A
6719770	12/2003	Laufer et al.	N/A	N/A
6719787	12/2003	Cox	N/A	N/A
6719788	12/2003	Cox	N/A	N/A
6719789	12/2003	Cox	N/A	N/A
6723116	12/2003	Taheri	N/A	N/A
6723122	12/2003	Yang et al.	N/A	N/A
6726677	12/2003	Flaherty et al.	N/A	N/A
6729356	12/2003	Baker et al.	N/A	N/A
6730118	12/2003	Spenser et al.	N/A	N/A
6730121	12/2003	Ortiz et al.	N/A	N/A
6730377	12/2003	Wang	N/A	N/A
6733513	12/2003	Boyle et al.	N/A	N/A
6733525	12/2003	Yang et al.	N/A	N/A
6736827	12/2003	McAndrew et al.	N/A	N/A
6736839	12/2003	Cummings	N/A	N/A
6736845	12/2003	Marquez et al.	N/A	N/A
6736846	12/2003	Cox	N/A	N/A
6743252	12/2003	Bates et al.	N/A	N/A
6746464	12/2003	Makower	N/A	N/A
6752828	12/2003	Thornton	N/A	N/A
6755854	12/2003	Gillick et al.	N/A	N/A
6755855	12/2003	Yurek et al.	N/A	N/A
6758855	12/2003	Fulton, III et al.	N/A	N/A
6764503	12/2003	Ishimaru	N/A	N/A
6764509	12/2003	Chinn et al.	N/A	N/A
6767345	12/2003	St. Germain et al.	N/A	N/A
6767362	12/2003	Schreck	N/A	N/A
6769434	12/2003	Liddicoat et al.	N/A	N/A
6773454	12/2003	Wholey et al.	N/A	N/A

6773455	12/2003	Allen et al.	N/A	N/A
6773456	12/2003	Gordon et al.	N/A	N/A
6774278	12/2003	Ragheb et al.	N/A	N/A
6776791	12/2003	Stallings et al.	N/A	N/A
6786925	12/2003	Schoon et al.	N/A	N/A
6786929	12/2003	Gambale et al.	N/A	N/A
6790229	12/2003	Berrekouw	N/A	N/A
6790230	12/2003	Beyersdorf et al.	N/A	N/A
6790237	12/2003	Stinson	N/A	N/A
6792979	12/2003	Konya et al.	N/A	N/A
6797000	12/2003	Simpson et al.	N/A	N/A
6797002	12/2003	Spence et al.	N/A	N/A
6802319	12/2003	Stevens et al.	N/A	N/A
6802858	12/2003	Gambale et al.	N/A	N/A
6805711	12/2003	Quijano et al.	N/A	N/A
6808498	12/2003	Laroya et al.	N/A	N/A
6808504	12/2003	Schorgl et al.	N/A	N/A
6808529	12/2003	Fulkerson	N/A	N/A
6814746	12/2003	Thompson et al.	N/A	N/A
6814754	12/2003	Greenhalgh	N/A	N/A
6820676	12/2003	Palmaz et al.	N/A	N/A
6821211	12/2003	Otten et al.	N/A	N/A
6821297	12/2003	Snyders	N/A	N/A
6824041	12/2003	Grieder et al.	N/A	N/A
6824970	12/2003	Vyavahare et al.	N/A	N/A
6830568	12/2003	Kesten et al.	N/A	N/A
6830575	12/2003	Stenzel et al.	N/A	N/A
6830584	12/2003	Seguin	N/A	N/A
6830585	12/2003	Artof et al.	N/A	N/A
6830586	12/2003	Quijano et al.	N/A	N/A
6837901	12/2004	Rabkin et al.	N/A	N/A
6837902	12/2004	Nguyen et al.	N/A	N/A
6840957	12/2004	DiMatteo et al.	N/A	N/A
6843802	12/2004	Villalobos et al.	N/A	N/A
6846325	12/2004	Liddicoat	N/A	N/A
6849084	12/2004	Rabkin et al.	N/A	N/A
6849085	12/2004	Marton	N/A	N/A
6854467	12/2004	Boekstegers	N/A	N/A
6860898	12/2004	Stack et al.	N/A	N/A
6861211	12/2004	Levy et al.	N/A	N/A
6863668	12/2004	Gillespie et al.	N/A	N/A
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6863688	12/2004	Ralph et al.	N/A	N/A
6866650	12/2004	Stevens et al.	N/A	N/A
6866669	12/2004	Buzzard et al.	N/A	N/A
6872223	12/2004	Roberts et al.	N/A	N/A
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6875231	12/2004	Anduiza et al.	N/A	N/A
6881199	12/2004	Wilk et al.	N/A	N/A
6881220	12/2004	Edwin et al.	N/A	N/A

6883522	12/2004	Spence et al.	N/A	N/A
6887266	12/2004	Williams et al.	N/A	N/A
6890330	12/2004	Streeter et al.	N/A	N/A
6890340	12/2004	Duane	N/A	N/A
6893459	12/2004	Macoviak	N/A	N/A
6893460	12/2004	Spenser et al.	N/A	N/A
6896690	12/2004	Lambrecht et al.	N/A	N/A
6899704	12/2004	Sterman et al.	N/A	N/A
6905743	12/2004	Chen et al.	N/A	N/A
6908481	12/2004	Cribier	N/A	N/A
6911036	12/2004	Douk et al.	N/A	N/A
6911040	12/2004	Johnson et al.	N/A	N/A
6911043	12/2004	Myers et al.	N/A	N/A
6913021	12/2004	Knudson et al.	N/A	N/A
6913600	12/2004	Valley et al.	N/A	N/A
6916304	12/2004	Eno et al.	N/A	N/A
6920674	12/2004	Thornton	N/A	N/A
6920732	12/2004	Mårtensson	N/A	N/A
6926690	12/2004	Renati	N/A	N/A
6926732	12/2004	Derus et al.	N/A	N/A
6929009	12/2004	Makower et al.	N/A	N/A
6929011	12/2004	Knudson et al.	N/A	N/A
6929653	12/2004	Strecker	N/A	N/A
6936058	12/2004	Forde et al.	N/A	N/A
6936066	12/2004	Palmaz et al.	N/A	N/A
6936067	12/2004	Buchanan	N/A	N/A
6939352	12/2004	Buzzard et al.	N/A	N/A
6939359	12/2004	Tu et al.	N/A	N/A
6939365	12/2004	Fogarty et al.	N/A	N/A
6939370	12/2004	Hartley et al.	N/A	N/A
6942682	12/2004	Vrba et al.	N/A	N/A
6945949	12/2004	Wilk	N/A	N/A
6945997	12/2004	Huynh et al.	N/A	N/A
6949080	12/2004	Wolf et al.	N/A	N/A
6949118	12/2004	Kohler et al.	N/A	N/A
6951571	12/2004	Srivastava	N/A	N/A
6953332	12/2004	Kurk et al.	N/A	N/A
6953481	12/2004	Phelps et al.	N/A	N/A
6955175	12/2004	Stevens et al.	N/A	N/A
6955681	12/2004	Evans et al.	N/A	N/A
6964652	12/2004	Guiles et al.	N/A	N/A
6964673	12/2004	Tsugita et al.	N/A	N/A
6964676	12/2004	Gerberding et al.	N/A	N/A
6969395	12/2004	Eskuri	N/A	N/A
6972025	12/2004	Wasdyke	N/A	N/A
6972029	12/2004	Mayrhofer et al.	N/A	N/A
6974464	12/2004	Quijano et al.	N/A	N/A
6974474	12/2004	Pavcnik et al.	N/A	N/A
6974476	12/2004	McGuckin, Jr. et al.	N/A	N/A
6976990	12/2004	Mowry	N/A	N/A

6979350	12/2005	Moll et al.	N/A	N/A
6984242	12/2005	Campbell et al.	N/A	N/A
6984244	12/2005	Perez et al.	N/A	N/A
6986742	12/2005	Hart et al.	N/A	N/A
6986784	12/2005	Weiser et al.	N/A	N/A
6988949	12/2005	Wang	N/A	N/A
6989027	12/2005	Allen et al.	N/A	N/A
6989028	12/2005	Lashinski et al.	N/A	N/A
6991649	12/2005	Sievers	N/A	N/A
7001425	12/2005	McCullagh et al.	N/A	N/A
7004176	12/2005	Lau	N/A	N/A
7008397	12/2005	Tweden et al.	N/A	N/A
7011095	12/2005	Wolf et al.	N/A	N/A
7011681	12/2005	Vesely	N/A	N/A
7014655	12/2005	Barbarash et al.	N/A	N/A
7018401	12/2005	Hyodoh et al.	N/A	N/A
7018406	12/2005	Seguin et al.	N/A	N/A
7018408	12/2005	Bailey et al.	N/A	N/A
7022134	12/2005	Quijano et al.	N/A	N/A
7025773	12/2005	Gittings et al.	N/A	N/A
7025780	12/2005	Gabbay	N/A	N/A
7025791	12/2005	Levine et al.	N/A	N/A
7028692	12/2005	Sterman et al.	N/A	N/A
7037331	12/2005	Mitelberg et al.	N/A	N/A
7037333	12/2005	Myers et al.	N/A	N/A
7041128	12/2005	McGuckin, Jr. et al.	N/A	N/A
7041132	12/2005	Quijano et al.	N/A	N/A
7044966	12/2005	Svanidze et al.	N/A	N/A
7048014	12/2005	Hyodoh et al.	N/A	N/A
7048757	12/2005	Shaknovich	N/A	N/A
7050276	12/2005	Nishiyama	N/A	N/A
7074236	12/2005	Rabkin et al.	N/A	N/A
7078163	12/2005	Torrianni	N/A	N/A
7081132	12/2005	Cook et al.	N/A	N/A
7097658	12/2005	Oktay	N/A	N/A
7097659	12/2005	Woolfson et al.	N/A	N/A
7101396	12/2005	Artof et al.	N/A	N/A
7105016	12/2005	Shiu et al.	N/A	N/A
7108715	12/2005	Lawrence-Brown et al.	N/A	N/A
7115141	12/2005	Menz et al.	N/A	N/A
7118585	12/2005	Addis	N/A	N/A
7122020	12/2005	Mogul	N/A	N/A
7125418	12/2005	Duran et al.	N/A	N/A
7128759	12/2005	Osborne et al.	N/A	N/A
7137184	12/2005	Schreck	N/A	N/A
7141063	12/2005	White et al.	N/A	N/A
7141064	12/2005	Scott et al.	N/A	N/A
7143312	12/2005	Wang et al.	N/A	N/A
7147662	12/2005	Pollock et al.	N/A	N/A

7147663	12/2005	Berg et al.	N/A	N/A
7153324	12/2005	Case et al.	N/A	N/A
7160319	12/2006	Chouinard et al.	N/A	N/A
7163556	12/2006	Xie et al.	N/A	N/A
7166097	12/2006	Barbut	N/A	N/A
7175652	12/2006	Cook et al.	N/A	N/A
7175653	12/2006	Gaber	N/A	N/A
7175654	12/2006	Bonsignore et al.	N/A	N/A
7175656	12/2006	Khairkhahan	N/A	N/A
7179290	12/2006	Cao	N/A	N/A
7186265	12/2006	Sharkawy et al.	N/A	N/A
7189258	12/2006	Johnson et al.	N/A	N/A
7189259	12/2006	Simionescu et al.	N/A	N/A
7191018	12/2006	Gielen et al.	N/A	N/A
7191406	12/2006	Barber et al.	N/A	N/A
7195641	12/2006	Palmaz et al.	N/A	N/A
7198646	12/2006	Figulla et al.	N/A	N/A
7201761	12/2006	Woolfson et al.	N/A	N/A
7201772	12/2006	Schwammenthal et al.	N/A	N/A
7214344	12/2006	Carpentier et al.	N/A	N/A
7217287	12/2006	Wilson et al.	N/A	N/A
7235092	12/2006	Banas et al.	N/A	N/A
7235093	12/2006	Gregorich	N/A	N/A
7238200	12/2006	Lee et al.	N/A	N/A
7241257	12/2006	Ainsworth et al.	N/A	N/A
7252682	12/2006	Seguin	N/A	N/A
7258696	12/2006	Rabkin et al.	N/A	N/A
7258891	12/2006	Pacetti et al.	N/A	N/A
7261732	12/2006	Justino	N/A	N/A
7264632	12/2006	Wright et al.	N/A	N/A
7267686	12/2006	DiMatteo et al.	N/A	N/A
7276078	12/2006	Spenser et al.	N/A	N/A
7276084	12/2006	Yang et al.	N/A	N/A
7285130	12/2006	Austin	N/A	N/A
7297150	12/2006	Cartledge et al.	N/A	N/A
7300457	12/2006	Palmaz	N/A	N/A
7300463	12/2006	Liddicoat	N/A	N/A
7311730	12/2006	Gabbay	N/A	N/A
7314449	12/2007	Pfeiffer et al.	N/A	N/A
7314485	12/2007	Mathis	N/A	N/A
7314880	12/2007	Chang et al.	N/A	N/A
7316706	12/2007	Bloom et al.	N/A	N/A
7316712	12/2007	Peredo	N/A	N/A
7317005	12/2007	Hoekstra et al.	N/A	N/A
7317942	12/2007	Brown	N/A	N/A
7317950	12/2007	Lee	N/A	N/A
7318278	12/2007	Zhang et al.	N/A	N/A
7318998	12/2007	Goldstein et al.	N/A	N/A
7319096	12/2007	Malm et al.	N/A	N/A

7320692	12/2007	Bender et al.	N/A	N/A
7320704	12/2007	Lashinski et al.	N/A	N/A
7320705	12/2007	Quintessenza	N/A	N/A
7320706	12/2007	Al-Najjar	N/A	N/A
7322932	12/2007	Xie et al.	N/A	N/A
7323006	12/2007	Andreas et al.	N/A	N/A
7323066	12/2007	Budron	N/A	N/A
7326174	12/2007	Cox et al.	N/A	N/A
7326219	12/2007	Mowry et al.	N/A	N/A
7326236	12/2007	Andreas et al.	N/A	N/A
7327862	12/2007	Murphy et al.	N/A	N/A
7329278	12/2007	Seguin et al.	N/A	N/A
7329279	12/2007	Haug et al.	N/A	N/A
7329280	12/2007	Bolling et al.	N/A	N/A
7329777	12/2007	Harter et al.	N/A	N/A
7331991	12/2007	Kheradvar et al.	N/A	N/A
7331993	12/2007	White	N/A	N/A
7333643	12/2007	Murphy et al.	N/A	N/A
7335158	12/2007	Taylor	N/A	N/A
7335213	12/2007	Hyde et al.	N/A	N/A
7335218	12/2007	Wilson et al.	N/A	N/A
7335490	12/2007	Van Gilst et al.	N/A	N/A
7338484	12/2007	Schoon et al.	N/A	N/A
7338520	12/2007	Bailey et al.	N/A	N/A
7361189	12/2007	Case et al.	N/A	N/A
7361190	12/2007	Shaouliau et al.	N/A	N/A
7364588	12/2007	Mathis et al.	N/A	N/A
7371258	12/2007	Woo et al.	N/A	N/A
7374560	12/2007	Ressemann et al.	N/A	N/A
7374571	12/2007	Pease et al.	N/A	N/A
7377895	12/2007	Spence et al.	N/A	N/A
7377938	12/2007	Sarac et al.	N/A	N/A
7377940	12/2007	Ryan et al.	N/A	N/A
7381210	12/2007	Zarbatany et al.	N/A	N/A
7381216	12/2007	Buzzard et al.	N/A	N/A
7381218	12/2007	Schreck	N/A	N/A
7381219	12/2007	Salahieh et al.	N/A	N/A
7381220	12/2007	Macoviak et al.	N/A	N/A
7384411	12/2007	Condado	N/A	N/A
7387640	12/2007	Cummings	N/A	N/A
7389874	12/2007	Quest et al.	N/A	N/A
7390325	12/2007	Wang et al.	N/A	N/A
7393358	12/2007	Malewicz	N/A	N/A
7393360	12/2007	Spenser et al.	N/A	N/A
7396364	12/2007	Moaddeb et al.	N/A	N/A
7399315	12/2007	Iobbi	N/A	N/A
7402171	12/2007	Osborne et al.	N/A	N/A
7404792	12/2007	Spence et al.	N/A	N/A
7404793	12/2007	Lau et al.	N/A	N/A
7405259	12/2007	Frye et al.	N/A	N/A

7410499	12/2007	Bicer	N/A	N/A
7412274	12/2007	Mejia	N/A	N/A
7412290	12/2007	Janke et al.	N/A	N/A
7415861	12/2007	Sokel	N/A	N/A
7416530	12/2007	Turner et al.	N/A	N/A
7422603	12/2007	Lane	N/A	N/A
7422606	12/2007	Ung-Chhun et al.	N/A	N/A
7423032	12/2007	Ozaki et al.	N/A	N/A
7426413	12/2007	Balczewski et al.	N/A	N/A
7427279	12/2007	Frazier et al.	N/A	N/A
7427287	12/2007	Turovskiy et al.	N/A	N/A
7427291	12/2007	Liddicoat et al.	N/A	N/A
7429269	12/2007	Schwammenthal et al.	N/A	N/A
7430448	12/2007	Zimmer et al.	N/A	N/A
7430484	12/2007	Ohara	N/A	N/A
7431691	12/2007	Wilk	N/A	N/A
7431733	12/2007	Knight	N/A	N/A
7435059	12/2007	Smith et al.	N/A	N/A
7435257	12/2007	Lashinski et al.	N/A	N/A
7442204	12/2007	Schwammenthal et al.	N/A	N/A
RE40570	12/2007	Carpentier et al.	N/A	N/A
7445630	12/2007	Lashinski et al.	N/A	N/A
7445631	12/2007	Salahieh et al.	N/A	N/A
7445632	12/2007	McGuckin, Jr. et al.	N/A	N/A
7452371	12/2007	Pavcnik et al.	N/A	N/A
7455689	12/2007	Johnson	N/A	N/A
7462156	12/2007	Mitrev	N/A	N/A
7462184	12/2007	Worley et al.	N/A	N/A
7462191	12/2007	Spenser et al.	N/A	N/A
7468050	12/2007	Kantrowitz	N/A	N/A
7470282	12/2007	Shelso	623/1.12	A61F 2/95
7470284	12/2007	Lambrecht et al.	N/A	N/A
7470285	12/2007	Nugent et al.	N/A	N/A
7473271	12/2008	Gunderson	N/A	N/A
7473275	12/2008	Marquez	N/A	N/A
7473417	12/2008	Zeltinger et al.	N/A	N/A
7476196	12/2008	Spence et al.	N/A	N/A
7476199	12/2008	Spence et al.	N/A	N/A
7476200	12/2008	Tal	N/A	N/A
7476244	12/2008	Buzzard et al.	N/A	N/A
7481838	12/2008	Carpentier et al.	N/A	N/A
7485088	12/2008	Murphy et al.	N/A	N/A
7485143	12/2008	Webler et al.	N/A	N/A
7488346	12/2008	Navia	N/A	N/A
7491232	12/2008	Bolduc et al.	N/A	N/A
7493869	12/2008	Foster et al.	N/A	N/A
7497824	12/2008	Taylor	N/A	N/A

7500949	12/2008	Gottlieb et al.	N/A	N/A
7500989	12/2008	Solem et al.	N/A	N/A
7503929	12/2008	Johnson et al.	N/A	N/A
7503930	12/2008	Sharkawy et al.	N/A	N/A
7507199	12/2008	Wang et al.	N/A	N/A
7510572	12/2008	Gabbay	N/A	N/A
7510574	12/2008	Le et al.	N/A	N/A
7510575	12/2008	Spenser et al.	N/A	N/A
7510577	12/2008	Moaddeb et al.	N/A	N/A
7513863	12/2008	Bolling et al.	N/A	N/A
7513909	12/2008	Lane et al.	N/A	N/A
7522950	12/2008	Fuimaono et al.	N/A	N/A
7524330	12/2008	Berrekouw	N/A	N/A
7530253	12/2008	Spenser et al.	N/A	N/A
7530995	12/2008	Quijano et al.	N/A	N/A
7534261	12/2008	Friedman	N/A	N/A
7544206	12/2008	Cohn	N/A	N/A
7547322	12/2008	Sarac et al.	N/A	N/A
7553324	12/2008	Andreas et al.	N/A	N/A
7556386	12/2008	Smith	N/A	N/A
7556646	12/2008	Yang et al.	N/A	N/A
7569071	12/2008	Haverkost et al.	N/A	N/A
7578828	12/2008	Gittings et al.	N/A	N/A
7585321	12/2008	Cribier	N/A	N/A
7591848	12/2008	Allen	N/A	N/A
7594974	12/2008	Cali et al.	N/A	N/A
7601159	12/2008	Ewers et al.	N/A	N/A
7601195	12/2008	Ichikawa	N/A	N/A
7608099	12/2008	Johnson et al.	N/A	N/A
7611534	12/2008	Kapadia et al.	N/A	N/A
7618446	12/2008	Andersen et al.	N/A	N/A
7622276	12/2008	Cunanan et al.	N/A	N/A
7625403	12/2008	Krivoruchko	N/A	N/A
7628802	12/2008	White et al.	N/A	N/A
7628803	12/2008	Pavcnik et al.	N/A	N/A
7632296	12/2008	Malewicz	N/A	N/A
7632298	12/2008	Hijlkema et al.	N/A	N/A
7635386	12/2008	Gammie	N/A	N/A
7641687	12/2009	Chinn et al.	N/A	N/A
7651519	12/2009	Dittman	N/A	N/A
7655034	12/2009	Mitchell et al.	N/A	N/A
7674282	12/2009	Wu et al.	N/A	N/A
7682390	12/2009	Seguin	N/A	N/A
7704222	12/2009	Wilk et al.	N/A	N/A
7704277	12/2009	Zakay et al.	N/A	N/A
7708775	12/2009	Rowe et al.	N/A	N/A
7712606	12/2009	Salahieh et al.	N/A	N/A
7715903	12/2009	Hartley et al.	N/A	N/A
7717955	12/2009	Lane et al.	N/A	N/A
7722638	12/2009	Deyette, Jr. et al.	N/A	N/A

7722662	12/2009	Steinke et al.	N/A	N/A
7722666	12/2009	Lafontaine	N/A	N/A
7722671	12/2009	Carlyle et al.	N/A	N/A
7731742	12/2009	Schlick et al.	N/A	N/A
7735493	12/2009	van der Burg et al.	N/A	N/A
7736327	12/2009	Wilk et al.	N/A	N/A
7736388	12/2009	Goldfarb et al.	N/A	N/A
7743481	12/2009	Lafont et al.	N/A	N/A
7748389	12/2009	Salahieh et al.	N/A	N/A
7758625	12/2009	Wu et al.	N/A	N/A
7763065	12/2009	Schmid et al.	N/A	N/A
7771463	12/2009	Ton et al.	N/A	N/A
7771467	12/2009	Svensson	N/A	N/A
7776083	12/2009	Vesely	N/A	N/A
7780725	12/2009	Haug et al.	N/A	N/A
7780726	12/2009	Seguin	N/A	N/A
7785360	12/2009	Freitag	N/A	N/A
7794487	12/2009	Majercak et al.	N/A	N/A
7799046	12/2009	White et al.	N/A	N/A
7799065	12/2009	Pappas	N/A	N/A
7803185	12/2009	Gabbay	N/A	N/A
7806919	12/2009	Bloom et al.	N/A	N/A
7823267	12/2009	Bolduc	N/A	N/A
7824442	12/2009	Salahieh et al.	N/A	N/A
7824443	12/2009	Salahieh et al.	N/A	N/A
7833262	12/2009	McGuckin, Jr. et al.	N/A	N/A
7837727	12/2009	Goetz et al.	N/A	N/A
7846203	12/2009	Cribier	N/A	N/A
7846204	12/2009	Letac et al.	N/A	N/A
7854758	12/2009	Taheri	N/A	N/A
7857845	12/2009	Stacchino et al.	N/A	N/A
7862602	12/2010	Licata et al.	N/A	N/A
7867274	12/2010	Hill et al.	N/A	N/A
7887583	12/2010	Macoviak	N/A	N/A
7892276	12/2010	Stocker et al.	N/A	N/A
7892292	12/2010	Stack et al.	N/A	N/A
7896913	12/2010	Damm et al.	N/A	N/A
7896915	12/2010	Guyenot et al.	N/A	N/A
7914569	12/2010	Nguyen et al.	N/A	N/A
7914574	12/2010	Schmid et al.	N/A	N/A
7914575	12/2010	Guyenot et al.	N/A	N/A
7918880	12/2010	Austin	N/A	N/A
7927363	12/2010	Perouse	N/A	N/A
7938851	12/2010	Olson et al.	N/A	N/A
7947071	12/2010	Schmid et al.	N/A	N/A
7947075	12/2010	Goetz et al.	N/A	N/A
7951189	12/2010	Haverkost et al.	N/A	N/A
7959666	12/2010	Salahieh et al.	N/A	N/A
7959672	12/2010	Salahieh et al.	N/A	N/A
7967853	12/2010	Eidenschink et al.	N/A	N/A

7972359	12/2010	Kreidler	N/A	N/A
7972376	12/2010	Dove et al.	N/A	N/A
7972378	12/2010	Tabor et al.	N/A	N/A
7988724	12/2010	Salahieh et al.	N/A	N/A
7993386	12/2010	Elliott	N/A	N/A
8002824	12/2010	Jenson et al.	N/A	N/A
8002825	12/2010	Letac et al.	N/A	N/A
8012198	12/2010	Hill et al.	N/A	N/A
8021421	12/2010	Fogarty et al.	N/A	N/A
RE42818	12/2010	Cali et al.	N/A	N/A
RE42857	12/2010	Cali et al.	N/A	N/A
8038704	12/2010	Sherburne	N/A	N/A
8038709	12/2010	Palasis et al.	N/A	N/A
8043450	12/2010	Cali et al.	N/A	N/A
8048153	12/2010	Salahieh et al.	N/A	N/A
8052715	12/2010	Quinn et al.	N/A	N/A
8052749	12/2010	Salahieh et al.	N/A	N/A
8052750	12/2010	Tuval et al.	N/A	N/A
8057540	12/2010	Letac et al.	N/A	N/A
8062355	12/2010	Figulla et al.	N/A	N/A
8062536	12/2010	Liu et al.	N/A	N/A
8062537	12/2010	Tuominen et al.	N/A	N/A
8062749	12/2010	Shelestak et al.	N/A	N/A
8070799	12/2010	Righini et al.	N/A	N/A
8075641	12/2010	Aravanis et al.	N/A	N/A
8083788	12/2010	Acosta et al.	N/A	N/A
8092518	12/2011	Schreck	N/A	N/A
8092520	12/2011	Quadri	N/A	N/A
8092521	12/2011	Figulla et al.	N/A	N/A
8128676	12/2011	Cummings	N/A	N/A
8128681	12/2011	Shoemaker et al.	N/A	N/A
8133217	12/2011	Stokes et al.	N/A	N/A
8133270	12/2011	Kheradvar et al.	N/A	N/A
8136659	12/2011	Salahieh et al.	N/A	N/A
8137394	12/2011	Stocker et al.	N/A	N/A
8137398	12/2011	Tuval et al.	N/A	N/A
8147534	12/2011	Berez et al.	N/A	N/A
8157853	12/2011	Laske et al.	N/A	N/A
8167894	12/2011	Miles et al.	N/A	N/A
8172896	12/2011	McNamara et al.	N/A	N/A
8182528	12/2011	Salahieh et al.	N/A	N/A
8192351	12/2011	Fishler et al.	N/A	N/A
8206437	12/2011	Bonhoeffer et al.	N/A	N/A
8211107	12/2011	Parks et al.	N/A	N/A
8216174	12/2011	Wilk et al.	N/A	N/A
8216301	12/2011	Bonhoeffer et al.	N/A	N/A
8221493	12/2011	Boyle et al.	N/A	N/A
8226707	12/2011	White	N/A	N/A
8226710	12/2011	Nguyen et al.	N/A	N/A
8231670	12/2011	Salahieh et al.	N/A	N/A

8236049	12/2011	Rowe et al.	N/A	N/A
8236241	12/2011	Carpentier et al.	N/A	N/A
8246675	12/2011	Zegdi	N/A	N/A
8246678	12/2011	Salahieh et al.	N/A	N/A
8252051	12/2011	Chau et al.	N/A	N/A
8252052	12/2011	Salahieh et al.	N/A	N/A
8277500	12/2011	Schmid et al.	N/A	N/A
8287584	12/2011	Salahieh et al.	N/A	N/A
8303653	12/2011	Bonhoeffer et al.	N/A	N/A
8308798	12/2011	Pintor et al.	N/A	N/A
8317858	12/2011	Straubinger et al.	N/A	N/A
8323335	12/2011	Rowe et al.	N/A	N/A
8328868	12/2011	Paul et al.	N/A	N/A
8343136	12/2012	Howat et al.	N/A	N/A
8343213	12/2012	Salahieh et al.	N/A	N/A
8348995	12/2012	Tuval et al.	N/A	N/A
8348996	12/2012	Tuval et al.	N/A	N/A
8348999	12/2012	Kheradvar et al.	N/A	N/A
8357387	12/2012	Dove et al.	N/A	N/A
8366767	12/2012	Zhang	N/A	N/A
8372134	12/2012	Schlick et al.	N/A	N/A
8376865	12/2012	Forster et al.	N/A	N/A
8377117	12/2012	Keidar et al.	N/A	N/A
8382822	12/2012	Pavcnik et al.	N/A	N/A
8398704	12/2012	Straubinger et al.	N/A	N/A
8398708	12/2012	Meiri et al.	N/A	N/A
8403983	12/2012	Quadri et al.	N/A	N/A
8414641	12/2012	Stocker et al.	N/A	N/A
8414643	12/2012	Tuval et al.	N/A	N/A
8414644	12/2012	Quadri et al.	N/A	N/A
8414645	12/2012	Dwork et al.	N/A	N/A
8439961	12/2012	Jagger et al.	N/A	N/A
8445278	12/2012	Everaerts et al.	N/A	N/A
8460365	12/2012	Haverkost et al.	N/A	N/A
8465540	12/2012	Straubinger et al.	N/A	N/A
8468667	12/2012	Straubinger et al.	N/A	N/A
8470023	12/2012	Eidenschink et al.	N/A	N/A
8491650	12/2012	Wiemeyer et al.	N/A	N/A
8512394	12/2012	Schmid et al.	N/A	N/A
8512399	12/2012	Lafontaine	N/A	N/A
8512400	12/2012	Tran et al.	N/A	N/A
8512401	12/2012	Murray, III et al.	N/A	N/A
8523936	12/2012	Schmid et al.	N/A	N/A
8535368	12/2012	Headley, Jr. et al.	N/A	N/A
8540762	12/2012	Schmid et al.	N/A	N/A
8545547	12/2012	Schmid et al.	N/A	N/A
8545552	12/2012	Garrison et al.	N/A	N/A
8551160	12/2012	Figulla et al.	N/A	N/A
8556880	12/2012	Freyman et al.	N/A	N/A
8556966	12/2012	Jenson	N/A	N/A

8568475	12/2012	Nguyen et al.	N/A	N/A
8579936	12/2012	Abbott et al.	N/A	N/A
8579962	12/2012	Salahieh et al.	N/A	N/A
8579965	12/2012	Bonhoeffer et al.	N/A	N/A
8585756	12/2012	Bonhoeffer et al.	N/A	N/A
8585759	12/2012	Bumbalough	N/A	N/A
8591570	12/2012	Revuelta et al.	N/A	N/A
8597226	12/2012	Wilk et al.	N/A	N/A
8603159	12/2012	Seguin et al.	N/A	N/A
8603160	12/2012	Salahieh et al.	N/A	N/A
8617235	12/2012	Schmid et al.	N/A	N/A
8617236	12/2012	Paul et al.	N/A	N/A
8623074	12/2013	Ryan	N/A	N/A
8623075	12/2013	Murray, III et al.	N/A	N/A
8623076	12/2013	Salahieh et al.	N/A	N/A
8623078	12/2013	Salahieh et al.	N/A	N/A
8628562	12/2013	Cummings	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8647381	12/2013	Essinger et al.	N/A	N/A
8668733	12/2013	Haug et al.	N/A	N/A
8672997	12/2013	Drasler et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685077	12/2013	Laske et al.	N/A	N/A
8696743	12/2013	Holecek et al.	N/A	N/A
8721713	12/2013	Tower et al.	N/A	N/A
8721717	12/2013	Shoemaker et al.	N/A	N/A
8734508	12/2013	Hastings et al.	N/A	N/A
8758430	12/2013	Ferrari et al.	N/A	N/A
8764818	12/2013	Gregg	N/A	N/A
8778020	12/2013	Gregg et al.	N/A	N/A
8790395	12/2013	Straubinger et al.	N/A	N/A
8795305	12/2013	Martin et al.	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8808356	12/2013	Braido et al.	N/A	N/A
8808364	12/2013	Palasis et al.	N/A	N/A
RE45130	12/2013	Figulla et al.	N/A	N/A
8828078	12/2013	Salahieh et al.	N/A	N/A
8828079	12/2013	Thielen et al.	N/A	N/A
8840662	12/2013	Salahieh et al.	N/A	N/A
8840663	12/2013	Salahieh et al.	N/A	N/A
8845721	12/2013	Braido et al.	N/A	N/A
8851286	12/2013	Chang et al.	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858620	12/2013	Salahieh et al.	N/A	N/A
8894703	12/2013	Salahieh et al.	N/A	N/A
8932349	12/2014	Jenson et al.	N/A	N/A
8940014	12/2014	Gamarra et al.	N/A	N/A
8951243	12/2014	Crisostomo et al.	N/A	N/A
8951299	12/2014	Paul et al.	N/A	N/A
8956383	12/2014	Aklog et al.	N/A	N/A

8992608	12/2014	Haug et al.	N/A	N/A
8998976	12/2014	Gregg et al.	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A
9011521	12/2014	Haug et al.	N/A	N/A
9023099	12/2014	Duffy et al.	N/A	N/A
9028542	12/2014	Hill et al.	N/A	N/A
9039756	12/2014	White	N/A	N/A
9044318	12/2014	Straubinger et al.	N/A	N/A
9131926	12/2014	Crisostomo et al.	N/A	N/A
9138315	12/2014	Straubinger et al.	N/A	N/A
9149358	12/2014	Tabor et al.	N/A	N/A
9168130	12/2014	Straubinger et al.	N/A	N/A
9168131	12/2014	Yohanan et al.	N/A	N/A
9168136	12/2014	Yang et al.	N/A	N/A
RE45790	12/2014	Figulla et al.	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9186482	12/2014	Dorn	N/A	N/A
9211266	12/2014	Iwazawa et al.	N/A	N/A
9216082	12/2014	Von Segesser et al.	N/A	N/A
9248037	12/2015	Roeder et al.	N/A	N/A
9265608	12/2015	Miller et al.	N/A	N/A
9277991	12/2015	Salahieh et al.	N/A	N/A
9277993	12/2015	Gamarra et al.	N/A	N/A
9295551	12/2015	Straubinger et al.	N/A	N/A
9301840	12/2015	Nguyen et al.	N/A	N/A
9301843	12/2015	Richardson et al.	N/A	N/A
9308085	12/2015	Salahieh et al.	N/A	N/A
9320599	12/2015	Salahieh et al.	N/A	N/A
9326853	12/2015	Olson et al.	N/A	N/A
9358106	12/2015	Salahieh et al.	N/A	N/A
9358110	12/2015	Paul et al.	N/A	N/A
9370419	12/2015	Hill et al.	N/A	N/A
9370421	12/2015	Crisostomo et al.	N/A	N/A
9387076	12/2015	Paul et al.	N/A	N/A
9393094	12/2015	Salahieh et al.	N/A	N/A
9393113	12/2015	Salahieh et al.	N/A	N/A
9393114	12/2015	Sutton et al.	N/A	N/A
9393115	12/2015	Tabor et al.	N/A	N/A
9415567	12/2015	Sogard et al.	N/A	N/A
9421083	12/2015	Eidenschink et al.	N/A	N/A
9439759	12/2015	Straubinger et al.	N/A	N/A
9463084	12/2015	Stinson	N/A	N/A
9474598	12/2015	Gregg et al.	N/A	N/A
9474609	12/2015	Haverkost et al.	N/A	N/A
9492276	12/2015	Lee et al.	N/A	N/A
9510945	12/2015	Sutton et al.	N/A	N/A
9510947	12/2015	Straubinger et al.	N/A	N/A
9526609	12/2015	Salahieh et al.	N/A	N/A
9532872	12/2016	Salahieh et al.	N/A	N/A
9539091	12/2016	Yang et al.	N/A	N/A

9554924	12/2016	Schlick et al.	N/A	N/A
9597432	12/2016	Nakamura	N/A	N/A
9649212	12/2016	Fargahi	N/A	N/A
9717593	12/2016	Alkhatib et al.	N/A	N/A
9744031	12/2016	Girard et al.	N/A	N/A
D800908	12/2016	Hariton et al.	N/A	N/A
9775709	12/2016	Miller et al.	N/A	N/A
9788945	12/2016	Ottma et al.	N/A	N/A
9861476	12/2017	Salahieh et al.	N/A	N/A
9867694	12/2017	Girard et al.	N/A	N/A
9867699	12/2017	Straubinger et al.	N/A	N/A
9872768	12/2017	Paul et al.	N/A	N/A
9878127	12/2017	Damm et al.	N/A	N/A
9889002	12/2017	Bonhoeffer et al.	N/A	N/A
9901445	12/2017	Backus et al.	N/A	N/A
9949824	12/2017	Bonhoeffer et al.	N/A	N/A
9956075	12/2017	Salahieh et al.	N/A	N/A
9968761	12/2017	Brecker	N/A	N/A
9987133	12/2017	Straubinger et al.	N/A	N/A
10092324	12/2017	Gillespie et al.	N/A	N/A
10143552	12/2017	Wallace et al.	N/A	N/A
10154901	12/2017	Straubinger et al.	N/A	N/A
10321987	12/2018	Wang et al.	N/A	N/A
10363134	12/2018	Figulla et al.	N/A	N/A
10543084	12/2019	Guyenot et al.	N/A	N/A
10575947	12/2019	Straubinger et al.	N/A	N/A
10638918	12/2019	Atarot et al.	N/A	N/A
10702382	12/2019	Straubinger et al.	N/A	N/A
10709555	12/2019	Schreck et al.	N/A	N/A
10856978	12/2019	Straubinger et al.	N/A	N/A
10856987	12/2019	Cabiri et al.	N/A	N/A
11065138	12/2020	Schreck et al.	N/A	N/A
11147669	12/2020	Straubinger	N/A	A61F 2/966
11154398	12/2020	Straubinger et al.	N/A	N/A
11185405	12/2020	Girard et al.	N/A	N/A
11197754	12/2020	Saffari et al.	N/A	N/A
11266497	12/2021	Cao et al.	N/A	N/A
11911264	12/2023	Chau et al.	N/A	N/A
11951005	12/2023	Gross et al.	N/A	N/A
12121461	12/2023	Schreck et al.	N/A	N/A
12171658	12/2023	Chu et al.	N/A	N/A
12232957	12/2024	Straubinger et al.	N/A	N/A
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2002/0111665	12/2001	Lauterjung	N/A	N/A
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2002/0161392	12/2001	Dubrul	N/A	N/A
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2002/0161424	12/2001	Rapacki et al.	N/A	N/A
2002/0161426	12/2001	Iancea	N/A	N/A
2002/0165479	12/2001	Wilk	N/A	N/A
2002/0165576	12/2001	Boyle et al.	N/A	N/A
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2002/0173842	12/2001	Buchanan	N/A	N/A
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2002/0177840	12/2001	Farnholtz	N/A	N/A
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2003/0044315	12/2002	Boekstegers	N/A	N/A
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2004/0098022	12/2003	Barone	N/A	N/A
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2008/0039925	12/2007	Ishimaru et al.	N/A	N/A
2008/0039934	12/2007	Styrc	N/A	N/A
2008/0045921	12/2007	Anderson et al.	N/A	N/A
2008/0048656	12/2007	Tan et al.	N/A	N/A
2008/0065001	12/2007	Dinuucci et al.	N/A	N/A
2008/0065011	12/2007	Marchand et al.	N/A	N/A
2008/0065206	12/2007	Liddicoat	N/A	N/A
2008/0071361	12/2007	Tuval et al.	N/A	N/A
2008/0071362	12/2007	Tuval et al.	N/A	N/A
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2008/0071368	12/2007	Tuval et al.	N/A	N/A
2008/0071369	12/2007	Tuval et al.	N/A	N/A
2008/0077227	12/2007	Ouellette et al.	N/A	N/A

2008/0077234	12/2007	Styrc	N/A	N/A
2008/0077236	12/2007	Letac et al.	N/A	N/A
2008/0082165	12/2007	Wilson et al.	N/A	N/A
2008/0082166	12/2007	Styrc et al.	N/A	N/A
2008/0086205	12/2007	Gordy et al.	N/A	N/A
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2008/0147105	12/2007	Wilson et al.	N/A	N/A
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2008/0154355	12/2007	Benichou et al.	N/A	N/A
2008/0154356	12/2007	Obermiller et al.	N/A	N/A
2008/0161909	12/2007	Kheradvar et al.	N/A	N/A
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2008/0172119	12/2007	Yamasaki et al.	N/A	N/A
2008/0177381	12/2007	Navia et al.	N/A	N/A
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2008/0188928	12/2007	Salahieh et al.	N/A	N/A
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2008/0200977	12/2007	Paul et al.	N/A	N/A
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2008/0208327	12/2007	Rowe	N/A	N/A
2008/0208328	12/2007	Antocci et al.	N/A	N/A
2008/0208332	12/2007	Lamphere et al.	N/A	N/A
2008/0215143	12/2007	Seguin	N/A	N/A
2008/0215144	12/2007	Ryan et al.	N/A	N/A
2008/0221672	12/2007	Lamphere et al.	N/A	N/A
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2008/0228254	12/2007	Ryan	N/A	N/A
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2008/0234443	12/2007	Kiss et al.	N/A	N/A
2008/0234797	12/2007	Styrc	N/A	N/A
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2008/0243246	12/2007	Ryan et al.	N/A	N/A
2008/0255651	12/2007	Dwork	N/A	N/A
2008/0255660	12/2007	Guyenot et al.	N/A	N/A
2008/0255661	12/2007	Straubinger et al.	N/A	N/A
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2009/0248143	12/2008	Laham	N/A	N/A
2009/0259306	12/2008	Rowe	N/A	N/A
2009/0264759	12/2008	Byrd	N/A	N/A
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2010/0198346	12/2009	Keogh et al.	N/A	N/A
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2010/0249916	12/2009	Zhang	N/A	N/A
2010/0249917	12/2009	Zhang	N/A	N/A
2010/0249918	12/2009	Zhang	N/A	N/A
2010/0256723	12/2009	Murray	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/390,120, filed on Apr. 22, 2019, now U.S. Pat. No. 11,147,669, which is a continuation of U.S. application Ser. No. 15/407,560, filed on Jan. 17, 2017, now U.S. Pat. No. 10,307,251, which is a continuation of U.S. application Ser. No. 13/698,910, filed on Mar. 18, 2013, now U.S. Pat. No. 9,597,182, which is a national phase application under 35 U.S.C. § 371 of PCT/EP2011/002524, filed on May 20, 2011, which is a continuation-in-part of U.S. patent application Ser. No. 12/801,090, filed on May 20, 2010, now U.S. Pat. No. 10,856,978, which claims the benefit of priority to European Application No. 10163478.0 filed on May 20, 2010, the entire contents of each of which are incorporated herein by reference.

SPECIFICATION

(1) The present disclosure concerns a catheter system for introducing an expandable heart valve

stent into the body of a patient. The disclosure further concerns an insertion system comprising a catheter system and a handle for inserting an expandable heart valve stent into the body of a patient, as well as a medical device for treatment of a heart valve defect, in particular a heart valve failure or a heart valve stenosis in a patient, wherein the medical device has an insertion system and an expandable heart valve stent accommodated in the catheter tip of the insertion system.

(2) In medical technology, there has been an endeavor over a long period to close a heart valve defect, such as an aortic valve insufficiency or an aortic valve stenosis, non-surgically by means of a transarterial interventional access by catheter, thus technically without an operation. Various insertion systems and stent systems have been proposed, with different advantages and disadvantages, which in part can be introduced into the body of a patient transarterially by means of a catheter insertion system, though a specific system has not prevailed up to the present.

(3) The term used here “heart valve stenosis and/or heart valve insufficiency” shall generally be understood here as a congenital or acquired functional disorder of one or several heart valves. A valve defect of this type can affect each of the four heart valves, whereby the valves in the left ventricle (aortic and mitral valve) are certainly more frequently affected than those of the right heart (pulmonary and tricuspid valve). The functional disorder can result in narrowing (stenosis) or inability to close (insufficiency) or a combination of the two (combined cardiac defect).

(4) With all known interventional systems for implantation of heart valve prosthesis, an expandable stent system is moved transarterially to an insufficient heart valve. A stent system of this type consists, for example, of a self-expanding or balloon-expanding anchoring support (also termed “heart valve stent” or “stent” in the following), to which the actual heart valve prosthesis is fastened, preferably at the distal retaining region of the anchoring support.

(5) In the medical devices previously known from the state-of-the-art, however, it has become apparent that the implantation procedure of a stent system to which the heart valve prosthesis is attached is relatively complicated, difficult and expensive. Apart from the complicated implantation of the heart valve prosthesis as a replacement for an insufficient native heart valve, there is the fundamental risk of incorrect positioning of the stent or heart valve prosthesis with the medical devices used up to the present, which cannot be corrected without more extensive operative intervention.

(6) The problem addressed by the present disclosure is the fact that medical technology does not currently offer any insertion system in particular for transarterial or transfemoral implantation of a self- or balloon-expandable heart valve stent with a heart valve prosthesis attached to it in which, on the one hand, the insertion system enables a minimally invasive implantation of the heart valve prosthesis in a predictable manner and, on the other, dispensing with the need to use a heart-lung machine during the operation on the anaesthetized patient. Consequently the operative intervention can be designed to be especially cost-effective and, in particular, to reduce the physical and mental stress on the patient. In particular, there is a lack of a medical device for implantation of heart valve prostheses that can also be used for patients on whom, due to their age, an operation cannot be carried out without the aid of a heart-lung machine.

(7) Because of the increasing number of patients requiring treatment, there is also a growing need for an insertion system with which a minimally invasive intervention can be made on a patient for treatment of a heart valve stenosis and/or heart valve insufficiency in a precisely predictable way, whereby the success of the operation is in particular no longer significantly dependent on the skill and experience of the heart surgeon or radiologist carrying out the treatment.

(8) This situation also applies to operations in which heart valve prostheses with stent systems are implanted with the aid of a so-called balloon catheter system.

(9) It is also regarded as problematic that, when using systems already known from the state-of-the-art by means of which a heart valve prosthesis can be implanted in the body of the patient with minimal invasiveness, incorrect positioning of the heart valve prosthesis or the associated heart valve stent can frequently only be avoided when the heart surgeon or radiologist is especially

experienced. It is indeed known, for example, to insert a heart valve stent with a heart valve prosthesis attached to it into the body of a patient as far as the heart via the aorta, whereby self-expansion or balloon-expansion of the heart valve stent is initiated by external manipulation when the implantation location is reached, which should lead to a secure anchorage and precise positioning of the heart valve prosthesis; such heart valve stents cannot usually be removed in a simple way, however, and their position cannot usually be corrected once the stent has expanded. (10) Accordingly, there is basically a risk with the known systems that if, for example, the self-expansion or balloon-expansion of the heart valve stent with the attached heart valve prosthesis is initiated in a non-optimum position, due to a slip by the doctor carrying out the treatment or other technical circumstances such as stent foreshortening, this position can only be corrected appropriately by means of a major, in particular operative, intervention, which must frequently be carried out on the open heart.

(11) For example, a heart valve stent for heart valve prosthesis is described in document WO 2004/019825 A1. With this heart valve stent, distal-end support arches or hoops and positioning arches or hoops are provided, which can be inserted into the pockets of the native heart valve of a patient so that the heart valve stent can be positioned by means of the support hoops. Additional so-called commissural hoops can also be formed on the known heart valve stent which, together with the support arches, clamp parts of the old heart valve once the stent has unfolded to that the stent can be positioned and anchored as a result of this clamping action.

(12) Although the support arches provided on the anchoring stent enable improved positioning of the heart valve prosthesis to be implanted, there is nevertheless still a risk of incorrect implantation and of the heart valve prosthesis being incapable of functioning correctly or functioning but unsatisfactorily. For example, it may be found during the intervention that the heart valve prosthesis or the heart valve stent is not optimally dimensioned for the patient. In such cases, even if only the respective distal support or positioning arches of the stent are in their expanded state, removal (explantation) or repositioning of the heart valve stent with the heart valve prosthesis is no longer possible and there exists an increased mortality risk for the particular patient.

(13) The aortic arch in the human body represents a further problem for such interventions, since it has to be accessed during insertion through the aorta. When this is done, the catheter tip and the respective catheter must undergo a change of direction of approximately 180° over a relatively small radius, usually about 50 mm, without causing injury or damage to the vessel wall.

(14) The objective of the disclosure is to propose a catheter system for introducing an expandable heart valve stent into the body of a patient and for positioning the stent at a desired implantation site, wherein the catheter system is designed to enable the implantation of a heart valve prosthesis attached to a heart valve stent in the optimum implantation location in a sequence of events defined before the intervention.

(15) Secondly, the objective is to propose a medical device for treatment of a heart valve stenosis and/or heart valve insufficiency, comprising a catheter system and an expandable heart valve stent mounted in the catheter tip of the insertion system and which is designed to reduce the risk to the patient on implantation of the heart valve prosthesis.

(16) In accordance with a preferred embodiment, the present disclosure provides a catheter system for introducing an expandable heart valve stent into the body of a patient, the catheter system comprising a catheter tip and a catheter shaft. The catheter tip of the catheter system has a seat portion for accommodating the stent to be introduced into the patient's body in its collapsed state. The catheter system has further a stent holder for realisably fixing the stent to the catheter tip. The seat portion of the catheter tip is constituted by a first sleeve-shaped member and a second sleeve-shaped member, said sleeve-shaped members being moveable relative to each other as well as relative to the stent holder of the catheter tip. The catheter shaft comprises first force transmitting means, second force transmitting means and guiding means. The distal end section of the first force transmitting means is connected to the first sleeve-shaped member of the catheter tip and the

proximal end section of the first force transmitting means is connectable to a first operating means of a handle. The distal end section of the second force transmitting means is connected to the second sleeve-shaped member of the catheter tip and the proximal end section of the second force transmitting means is connectable to a second operating means of the handle.

(17) Preferably, the cross-section of second sleeve-shaped member of the catheter tip is equal to or less than the cross-section of the first sleeve-shaped member of the catheter tip. In case the cross-section of second sleeve-shaped member of the catheter tip is less than the cross-section of the first sleeve-shaped member, the second sleeve-shaped member is at least partly accommodatable within the first sleeve-shaped member in a telescopic manner. This may allow minimizing the cross-section of catheter tip. At the same time, an expandable heart valve stent may be released from the catheter tip of the catheter system in a step-wise manner. In case the cross-section of second sleeve-shaped member of the catheter tip is less than the cross-section of the first sleeve-shaped member, the second sleeve-shaped member and the first sleeve-shaped member—once brought together—can reside on an internal support structure, e.g. a cylindrical insert, resulting in a step and gap free transition.

(18) According to one aspect of the present disclosure, the catheter system comprises guiding means having a guiding tube with a passageway extending there between. The guiding means serves for guiding of the catheter shaft has a distal end, a proximal end and a passageway extending there between. The first and second force transmitting means are at least partly received within this passageway such as to be moveable relative to the guiding means. The guiding tube of the guiding means has a length such that the distal end of the guiding means terminates proximal to the catheter tip of the catheter system. Moreover, guiding tube has a cross-section less than the cross-section of the catheter tip.

(19) According to another aspect of the present disclosure, the catheter system further comprises a guide wire suited for guiding the catheter tip of the catheter system to an implantation site. The guide wire is designed to be advanced into a patient's vasculature independently from the catheter system and, in particular, independently from the catheter tip of the catheter system.

(20) In accordance with another preferred embodiment, an insertion system for inserting an expandable heart valve stent is disclosed.

(21) Whilst the term “vascular” refers to the blood vessels of the patient's body including both veins and arteries, in a preferred embodiment, the insertion system is for transarterial delivery using the arteries, although it is conceivable that in other embodiments transvenous delivery via a vein could be used.

(22) In particular, the vascular insertion system comprises a catheter system with a catheter tip, a catheter shaft and a handle. The catheter tip has a seat portion for accommodating a stent to be inserted in its collapsed state and a stent holder for releasably fixing the stent. The proximal end of the catheter system is attached to the handle and the distal end is attached to the catheter tip. The catheter system comprises the catheter shaft for connecting the catheter tip to the handle of the insertion system, the distal end section of the catheter shaft being flexible enough such that the catheter tip and the distal end section of the catheter shaft may be easily navigated through the anatomy and especially through the aortic arch during insertion through the aorta of the patient.

(23) The handle has at least one first and one second operating means with which the catheter tip of the insertion system may be appropriately manipulated so that an expandable stent housed in the catheter tip may be released from the catheter tip in steps or in a defined or definable sequence of events.

(24) The catheter tip of the catheter system and at least the distal part of the catheter shaft are typically inserted into the femoral artery and moved up the descending thoracic aorta until the catheter tip is positioned in the ascending aorta. The proximal end of the catheter shaft together with the handle attached thereto remains outside of the patient's body.

(25) In accordance with a preferred embodiment, the catheter tip has first and second housing

portions termed “sleeve-shaped members” in the following, that may be manipulated with the handle. These sleeve-shaped members are used for accommodating specific portions of the stent. The first sleeve-shaped member is used for accommodating first functional components of the stent, for example retaining hoops of the stent (or alternatively positioning hoops of the stent), while the second sleeve-shaped member is used for accommodating the second functional components of the stent, for example, positioning hoops of the stent (or alternatively for accommodating retaining hoops of the stent).

(26) In relation to the handle provided for the insertion system, it is preferably provided that, on one hand, the first operating means cooperate with the first sleeve-shaped member of the catheter tip so that, on actuation of the first operating means, a previously definable longitudinal displacement of the first sleeve-shaped member may be effected relative to the stent holder and the guiding tube of the catheter shaft. On the other hand, the second operating means cooperates with the second sleeve-shaped member of the catheter tip so that a previously definable longitudinal displacement of the second sleeve-shaped member may be affected relative to the stent holder and the guiding tube of the catheter shaft.

(27) The cross-section of the second sleeve-shaped member is identical to the cross-section of the first sleeve-shaped member such that the sleeve-shaped members can completely enclose a stent accommodated in the catheter tip without a gap between the first and second sleeve-shaped members thereby providing a catheter tip having an atraumatic shape. In addition, the first and second sleeve-shaped members are movable relative to each other and relative to the stent holder.

(28) For this purpose, first force transmitting means with a distal end section connected to the first sleeve-shaped member and a proximal end section connected to first operating means of the handle are provided. In addition, second force transmitting means with a distal end section connected to the second sleeve-shaped member and a proximal end section connected to second operating means of the handle are provided. When manipulating the first and/or second operating means of the handle, the first and/or second sleeve-shaped members may be moved relative to each other and relative to the stent holder.

(29) In accordance with the preferred embodiment, the first force transmitting means is constituted by a first catheter tube defining a first lumen and the second force transmitting means is constituted by a second catheter tube defining a second lumen. The second catheter tube has a cross-section less than the cross-section of the first catheter tube. The first catheter tube is disposed concentrically and coaxially with the second catheter tube and the second catheter tube is received within the first lumen defined by the first catheter tube.

(30) Contrary to the first and second sleeve-shaped members of the catheter tip, however, the stent holder of the catheter tip is not moveable relative to the handle of the insertion system. Rather, the stent holder is connected to the handle by using a stent holder tube having a distal end connected to the stent holder and a proximal end connected to a body of the handle. The stent holder tube has a cross-section less than the cross-section of the first catheter tube. In particular, the first catheter tube is disposed concentrically and coaxially with both, the second catheter tube on the one hand and the stent holder tube on the other hand. Preferably, the stent holder tube has a cross-section less than the cross-section of the first catheter tube and greater than the cross-section of the second catheter tube such that the stent holder tube is received within the first lumen defined by the first catheter tube and the second catheter tube is received within a passageway defined by the stent holder tube. The passageway defined by the stent holder tube has a diameter sufficient to accommodate the second catheter tube such that the second catheter tube is moveable relative to the stent holder tube.

(31) The second lumen defined by the second catheter tube has a diameter sufficient to accommodate a guide wire. The second catheter tube is made from a rigid material including, for example, nitinol, stainless steel or a rigid plastic material. The material of the distal end section of the second catheter tube may have an increased flexibility compared to the material of the proximal

end section in order to allow the distal end section of the catheter shaft to pass the aortic arch during insertion of the catheter tip.

(32) The distal end section of the second catheter tube terminates in a soft catheter end tip having an atraumatic shape. The soft catheter end tip is provided with a channel aligned with the second lumen defined by the second catheter tube such that a guide wire accommodated within the second lumen of the second catheter tube may pass through the channel of the soft catheter end tip. The second sleeve-shaped member of the catheter tip is connected to the soft catheter end tip such that the opened end of the second sleeve-shaped member faces in the proximal direction opposite to the direction of the soft catheter end tip and to the second catheter tube.

(33) The stent holder tube is made of a rigid material, for example, a rigid plastic material, stainless steel or nitinol. The distal end of the stent holder tube terminates in the stent holder which is also made of a rigid material, for example, a rigid plastic material or stainless steel. The passageway defined by the stent holder tube is aligned with a channel which passes through the stent holder. In this way, the second catheter tube is accommodated in the passageway of the stent holder tube and the channel of the stent holder such as to be moveable relative to the stent holder tube and the stent holder. The stent holder tube is provided for connecting the stent holder to the handle. For this purpose, the stent holder tube has a distal end connected to the stent holder and a proximal end connected to a body of the handle.

(34) The first catheter tube is made of a bendable but inelastic material. For example, the first catheter tube may be at least partly made of a braided or non-braided catheter tube. Hence, the first catheter tube has a stiff braid reinforced body similar to the catheter body described in U.S. Pat. No. 4,665,604 which is incorporated herein by reference.

(35) The first catheter tube shall be adapted to transfer compression and tension forces from the first operating means of the handle to the first sleeve-shaped member of the catheter tip without overly changing of its total length. The distal end of the first catheter tube terminates at a flared section as the transition to the section defining the first sleeve-shaped member of the catheter tip. The flared section and the first sleeve-shaped member may be formed integrally and may be connected to the distal end section of the first catheter tube. Alternatively, the first sleeve-shaped member and the flared section of the first catheter tube may be all of the same material and originating from the same raw tube prior to a widening process so that the flared section and the first sleeve-shaped member are the same elements.

(36) The insertion system according to the preferred embodiment further comprises a guiding tube having a cross-section greater than the cross-section of the first catheter tube. The guiding tube defines a passageway and is disposed concentrically and coaxially with the first catheter tube, the stent holder tube and the second catheter tube such that the first catheter tube with the stent holder tube and the second catheter tube accommodated therein is at least partly accommodated within the passageway defined by the guiding tube, wherein the first catheter tube is moveable relative to the guiding tube. In particular, the guiding tube terminates proximal to the catheter tip wherein the cross-section of proximal end section of the guiding tube shall be substantially the same as or less than the cross-section of the flared section provided at the proximal end of the first catheter tube. The proximal end section of the guiding tube terminates distal to the handle. The proximal end section of the guiding tube may be detached/disconnected from the handle so that the handle as well as the first and second catheter tubes and the stent holder tube together with catheter tip may be moved relative to the guiding tube.

(37) The distal end of the guiding tube is formed such that the flared section provided at the distal end section of the first catheter tube may abut on the distal end of the guiding tube without abrupt transition. The guiding tube may be of a thin material such as to allow length deformation of the guiding tube upon transfer of compression and tension forces. The guiding tube material, however, shall have sufficient stiffness in order to mechanically avoid kinking of the flexible sections of the distal portion of the catheter shaft during insertion of the catheter tip.

(38) The proximal end of the guiding tube is releasably connectable to the body of the handle. In this way, the guiding tube may have a double-function:

(39) In case, the proximal end of the guiding tube is connected to the handle, the guiding tube serves as a distal extension of the body of the handle relative to which the first and second operating means are moveable for manipulating the first and second sleeve-shaped members of the catheter tip. Hence, position of the stent holder relative to the native heart valve of the patient may be changed by moving the guiding tube connected to the handle.

(40) In case, the proximal end of the guiding tube is not connected to the body of the handle, the guiding tube may serve as a portal for passing the catheter shaft of the catheter system into the patient's body from proximal of the catheter tip.

(41) In any case, the guiding tube has a length and is adapted such that the first catheter tube and the second catheter tube are moveable relative to each other and relative to the stent holder independent from any movement or activation of the guiding tube. In particular, the movement of the sleeve shaped members is independent from the presence or absence of the guiding tube. The length of the guiding tube is such that the sleeved shaped members and hence the first and second catheter tubes are moveable relative to each other and relative to the stent holder without interfering with the distal end of the guiding tube.

(42) An inlet may be provided at a proximal end section of the guiding tube for injection of fluids into the guiding tube. Furthermore, a check valve may be provided at the proximal end section of the guiding tube to prevent fluid from leaking out of the guiding tube.

(43) The guiding tube may have a length sufficient to protect the inner wall of the blood vessel through which the catheter tip passes. In addition, a separate introducer system (not belonging to the catheter system) may be provided. The introducer system then may serve as a portal for passing the complete catheter system from the catheter tip to the catheter shaft into the patient's body and up to the heart.

(44) In addition, the guiding tube reduces the compression force exerted on the first catheter tube that is inserted through the guiding tube. This increases manoeuvrability of the first catheter tube throughout the procedure in which the first catheter tube serves as force transmitting means for manipulating the first sleeve-shaped member of the catheter tip. A consequence thereof is that the frictional force acting on the first catheter tube is reduced compared with a catheter design which is not provided with a guiding tube. Moreover, moving the catheter tip after it has been advanced through the vascular system of a patient, is greatly improved while at the same time lowering the risk of injury of the patient.

(45) In accordance with the preferred embodiment, the guiding tube has a cross-section equal to or less than the cross-section of the catheter tip. In this regard, the guiding tube will have a length shorter than the length of the first and second catheter tubes such that the distal end of the guiding tube terminates proximal to the catheter tip. As will be appreciated, the guiding tube may not be removed from the catheter system in case the proximal end sections of the first and second catheter tube are connected to the respective operating means of a handle.

(46) The length of the guiding tube depends on the length of the first and second catheter tubes and will typically be between about 20 cm and 100 cm. Those skilled in the art will appreciate, however, that all dimensions provided herein are intended as examples only, and that the guiding tubes and catheter tubes of different dimensions may be substituted for a particular use. As already indicated, the first and second catheter tubes are moveable relative to each other and relative to the stent holder independent from the guiding tube. The movement of the sleeve shaped members is independent from the presence or absence of the guiding tube. In other words, the guiding tube does not serve for manipulating the sleeve-shaped members of the catheter tip. In particular, the guiding tube does not block the travel of the sleeve-shaped members.

(47) As will be appreciated, the guiding tube will be of a size, i.e. has an outer diameter, which will permit insertion in a patient's blood vessel (artery or vein) which is used for moving the stent

transarterially or via a vein to an insufficient heart valve.

(48) The guiding tube may be capable of traversing tortuous pathways in the body of the patient without kinking. The guiding tube may include an inner lubricious liner, an outer polymeric jacket, and a coil reinforcement between the inner and the outer layers. This guiding tube may provide favourable flexibility without kinking or compression. One or more radiopaque bands or markers may be incorporated within the guiding tubes material to allow precise location of the guiding tubes distal end for positioning accuracy. Those skilled in the art will appreciate that other known materials may also be suitable for a particular purpose.

(49) In an embodiment disclosed herein, the catheter tip and the catheter shaft proximally connected to the catheter tip may be inserted into the patient's body by using a guide wire. The guide wire serves for guiding the catheter tip of the catheter system to an implantation site. Once in position above the aortic valve the guide wire may then be removed. Alternatively, the guide wire remains in the patient's body during implantation of a heart valve prosthesis accommodated in the catheter tip. Then, the guide wire is removed together with the catheter from the patient's body.

(50) The guide wire is designed to be advanced into a patient's vasculature independently from the catheter tip and the catheter shaft proximally connected to the catheter tip. In other words, the catheter tip together with at least the distal part of the catheter shaft and the guide wire are advanced as single units through the vasculature of the patient, respectively. Once the guide wire is placed, the catheter tip and the catheter shaft proximally connected to the catheter tip can be advanced over the guide wire directly to the particular site in the patient's cardiovascular system.

(51) In accordance with the present invention, a guide wire is advanced through the patient's vascular system, its direction being controlled and fluoroscopically monitored by the surgeon, until its distal end is at the desired location. Preferably, the guide wire is very small in diameter, thereby not presenting any substantial obstruction to blood flow in the blood vessel. After inserting the guide wire, the catheter tip together with the catheter shaft proximally connected to the catheter tip are advanced over the guide wire with the wire being received in the second lumen which is defined by the second catheter tube of the catheter shaft. The guide wire thus simply and automatically guides the catheter tip of the catheter system directly to the intended region, without requiring difficult, time consuming manipulations.

(52) In a preferred embodiment of the present disclosure, the guide wire has a diameter less than the diameter of the second lumen defined by the second catheter tube. This allows that the guide wire may be at least partly received within the second lumen defined by the second catheter tube for guiding the catheter tip, at least partly disposed about the guide wire, to the implantation site.

(53) In a preferred embodiment of the invention, the second lumen defined by the second catheter tube of the catheter shaft has a minimum dimension which is just slightly greater than the diameter of the guide wire. The maximum cross-sectional dimension of the second lumen is substantially larger than the cross-section of the guide wire. Thus, when the guide wire is disposed within the second lumen there will be substantial voids through the second lumen, on opposite sides of the guide wire through which fluids may be administered to the patient and through which blood pressure measurements may be taken. Such fluids may be administered and pressure measurements may be taken without removing the guide wire at all. By way of example, the cross-section of the guide wire preferably is of the order of no more than about fifty percent of the cross-sectional area of the second lumen.

(54) In order to implant a heart valve prosthesis accommodated in the catheter tip, the catheter tip and the catheter shaft proximally connected to the catheter tip are advanced over the guide wire. As the tip of the guide wire terminates in the left ventricle of the heart, pushing the guide wire may contact the left ventricular apex.

(55) In order to avoid any damage of the left ventricular apex when the guide wire is inserted, and to avoid any injury or damage to the vessel wall when the guide wire is inserted (advanced) through a vessel, the guide wire preferably has a flexible bumper at the leading end of the

advancing guide wire, which minimizes the risk of trauma or injury to the delicate internal surfaces of the artery. The bumper is preferably highly flexible and with a smooth leading end. For example, the guide wire may terminate in a smoothly surfaced rounded tip, at the distal end of the guide wire. Alternatively, the distal end of the guide wire may have a j-hook shape or a hockey-stick shape, thereby reducing the risk of trauma.

(56) Since the guide wire and the catheter tip together with the catheter shaft proximally connected to the catheter tip are generally independently advanced into the vasculature, the guide wire must be sufficiently stiff throughout its length to prevent buckling. Furthermore, the guide wire shall have sufficient stiffness to track the delivery system (catheter tip and catheter shaft proximally connected to the catheter tip) around the aortic arch. On the other hand, at least the distal tip of the guide wire shall be soft enough to prevent puncture of the heart tissue.

(57) The guide wire may comprise a distal tip guide section and a proximal pull section. The pull section allowing for the tip guide section to be pulled out after final positioning of the catheter tip and having an optimal cross sectional area and size, is generally smaller than that of the tip guide section so as to assure minimum blood leakage at the insertion site. The tip guide section is capable of guiding the catheter through a patient's vasculature.

(58) The guide wire may, in an exemplary embodiment, be approximately 175 centimeters long so that it may be introduced through the femoral artery and have ample length to reach the patient's coronary region. The guide wire may include a small diameter main wire. This rotationally rigid main wire of the guide wire can be solid or tubular, as long as it is rigid torsionally so that it may transmit fully to the distal end a rotational motion imparted to the proximal end. The main wire has relatively little twist as its proximal end is rotated. Practically all rotation applied to the proximal end will be transmitted quickly to the very distal tip of the guide wire. Alternatively, the guide wire may be formed substantially from elongate helical springs.

(59) As already indicated, the aortic arch in the human body may represent a challenge for transfemoral implantation of a self- or balloon-expandable heart valve stent with a heart valve prosthesis attached to it, since it has to be accessed during insertion through the aorta. When this is done, the catheter tip and the catheter shaft proximally connected to the catheter tip must undergo a change of direction of approximately 180° over a relatively small radius, usually about 50 mm, without causing injury or damage to the vessel wall. For aiding the bending of the catheter tip and the catheter shaft proximally connected to the catheter tip when passing through the aortic arch and for supporting the catheter tip in accessing the ascending aorta, the guide wire may have a specific structure such as to make a U turn in the aortic arch. Hence, the guide wire may be programmed such that the guide wire takes a U-shape bend.

(60) In a preferred embodiment of the present disclosure, at least a distal section of the guide wire has a predefined curved configuration adapted to the curvature of the patient's aortic arch. The predefined curved configuration of at least the distal section of the guide wire is selected such as to push the catheter tip in the direction of the centre of the ascending aorta when the catheter tip is at least partly disposed about the distal section of the guide wire and transfemoral inserted into the patient's body.

(61) In this respect, the guide wire has a double-function: On the one hand, the guide wire serves for guiding the catheter tip of the catheter system to an implantation site. On the other hand, the guide wire serves for positioning the catheter tip in the centre of the ascending aorta when the catheter tip has accessed the ascending aorta. Then, positioning arches or hoops of the stent accommodated in the catheter tip may be easily inserted into the pockets of the native heart valve of a patient so that the heart valve stent can be easily positioned.

(62) In a preferred embodiment, at least the distal section of the guide wire exhibits a first predefinable shape before advancing the guide wire into the patient's vasculature and a second predefinable shape in the advanced state of said guide wire, wherein the second predefinable shape of the distal section of the guide wire corresponds to the predefined curved configuration of the

distal section of the guide wire. For achieving this the guide wire may consist at least partly of a shape memory material such that at least the distal section of the guide wire can transform from a temporary shape into a permanent shape under influence of an external stimulus, wherein the temporary shape of the distal section of the guide wire corresponds to the first shape and the permanent shape of the distal section of the guide wire corresponds to the second shape.

(63) A shape memory material, for example Nitinol, may be used as the material for at least the distal section of the guide wire. Such a shape memory material is preferably designed such that the guide wire can transform from a temporary shape into a permanent shape under the influence of an external stimulus. The temporary shape is thereby the first shape of the guide wire (i.e. the shape of the guide wire before inserting it into the patient's body), while the permanent shape is assumed in the second shape of the guide wire (i.e. in the inserted state of the guide wire). In particular, use of a shape memory material such as Nitinol, i.e. an equiatomic alloy of nickel and titanium, allows for a particularly gentle insertion procedure.

(64) It is conceivable of course that other shape memory materials, for example shape-memory polymers, are used as the material for at least the distal section of the guide wire. At least parts of the guide wire may be formed by using, for example, a polymer composite exhibiting a crystalline or semi-crystalline polymer network having crystalline switching segments. On the other hand, an amorphous polymer network having amorphous switching segments is also conceivable.

(65) When manufacturing the guide wire preferably made from a shape memory material, the permanent shape of the guide wire, i.e. the shape of the guide wire which is assumed in the inserted state of the guide wire, is formed. Once the desired shape has been formed, this shape is "fixed", this process being known as "programming". Programming may be effected by heating the guide wire, forming the guide wire into the desired shape and then cooling the guide wire. Programming may also be effected by forming and shaping the structure of the guide wire at lower temperature, this being known as "cold stretching." The permanent shape is thus saved, enabling the guide wire to be stored and implanted in a temporary, non-formed shape. If an external stimulus then acts on the stent structure, the shape memory effect is activated and the saved, permanent shape restored.

(66) A particularly preferred embodiment provides for the external stimulus to be a definable switching temperature. It is thus conceivable that the material of the guide wire needs to be heated to a higher temperature than the switching temperature in order to activate the shape memory effect and thus regenerate the saved permanent shape of the guide wire. A specific switching temperature can be preset by the relevant selection of the chemical composition of the shape memory material.

(67) It is particularly preferred to set the switching temperature to be in the range of between 10° C. and the patient's body temperature and preferably in the range of between 10° C. and room temperature (22° C.). Doing so is of advantage, especially with regard to the guide wire which needs to be inserted in a patient's body. Accordingly, all that needs to be ensured in this regard when inserting the guide wire is that the guide wire is warmed up to room temperature or the patient's body temperature (37° C.) at the site of implantation to activate the shape memory effect of the stent material.

(68) Alternatively, the guide wire may be made from another material (for example a platinum-tungsten alloy) which allows that the distal region of the guide wire can be bent manually by the surgeon and will retain its bent configuration when relaxed. This enables the guide wire to be controllably steered by rotation of the guide wire to direct the curved distal end selectively into the aortic arch and into the ascending aorta. Rotational control of the guide wire may be enhanced by bending the proximal end of the wire to form somewhat of a handle.

(69) In use, the surgeon may bend the distal region of the guide wire so that it will be biased toward and will assume somewhat of a curve when relaxed. When advanced through the patient's artery the degree of resilience at the distal region of the wire is such that the wire will straighten and follow the path of the artery quite easily. A progressively increased flexibility resulting from, for example, a continuous taper at the distal region of the guide wire may enhance the ability of the guide wire to

flex from the pre-bent biased curve and follow the path of the blood vessel.

(70) When the distal end of the pre-bent, biased guide wire is at the descending aorta proximal of the aortic arch, the surgeon can steer it into the aortic arch and thereafter into the ascending arch by rotation of the guide wire by manipulating it from the proximal end.

(71) Alternatively, the guide wire may be inserted into the patient's body by using a guide catheter. The guide catheter may comprise a guide catheter tube defining a lumen for receiving the guide wire. The guide catheter may serve for inserting the guide wire. Once the guide catheter is introduced through the femoral artery and the aortic arch and has reached the patient's aortic valve region, the guide wire is released from the guide catheter by removing the guide catheter whereas the guide wire remains in the patient's body. In this case, the guide wire exhibits its first predefinable shape before releasing the guide wire from the guide catheter and its second predefinable shape after releasing the guide wire from the guide catheter. As already indicated, the second predefinable shape of the distal section of the guide wire is selected such that the distal section of the guide wire pushes the catheter tip in the direction of the centre of the ascending aorta when the catheter tip is at least partly disposed about the distal section of the guide wire and transfemoral inserted into the patient's body.

(72) According to one aspect of the present disclosure, at least the distal region of the guide wire is at least partly formed from a material having a high radiopacity. A relatively high degree of radiopacity of the distal region of the guide wire enhances fluoroscopic imaging of the guide wire as it is advanced through the patient's artery.

(73) The procedure for using the guide wire in accordance with the present invention involves initial placement and location of the guide wire in the femoral artery and the aortic arch. Once the guide wire is in place the catheter tip with the catheter shaft of the catheter system then may be advanced over the guide wire to a point where the stent accommodated in the catheter tip is in the ascending aorta proximal to the native aortic heart valve. This can be verified fluoroscopically because of the highly radiopaque characteristic of the catheter tip and/or guide wire and also by injecting radiopaque dye through, for example, a lumen of the catheter system. For this reason, the catheter tip of the catheter system may be provided with radiopaque markers which also facilitate fluoroscopic monitoring of its progress and position.

(74) In order to treat a heart valve stenosis and/or heart valve insufficiency in a patient, a medical device is disclosed. The medical device comprises an insertion system and an expandable heart valve stent accommodated in the catheter tip of the insertion system. While it is accommodated in the catheter tip of the insertion system, the stent adopts a first previously definable configuration. Outside the catheter tip or in the implanted state, however, the stent exists in a second previously definable configuration. The first configuration of the stent corresponds to the folded-up state, while the stent exists in its expanded state in the second configuration.

(75) A heart valve stent is used with the medical device, as described for example in the European Patent Application No. 07 110 318 or in the European Patent Application No. 08 151 963. In a preferred embodiment of the medical device, a heart valve stent is accordingly used which exhibits the following: a first retaining region, to which a heart valve prosthesis can be attached; an opposing, second retaining region with at least one retaining element, for example in the form of retaining eyes or in the form of retaining heads, whereby at least one retaining element of the stent can be put in releasable engagement with the stent holder of the catheter tip forming part of the insertion system; at least one retaining hoop, to which a heart valve prosthesis can be fastened; and at least one and preferably three positioning hoops, which are designed to engage in pockets of the native heart valve in the implanted state of the stent, thus to enable automatic positioning of the stent in the aorta of the patient.

(76) In particular, an insertion system is proposed, with which an expandable heart valve stent with a heart valve prosthesis attached to this stent can be advanced to the implantation site in a particularly simple way, for example via the aorta of a patient being treated (transarterially or

transfemorally). Preferably, during transarterial or transfemoral access by the catheter system, the whole free cross-section available within the aorta is not completely filled up, since the catheter tip provided at the distal end region of the catheter system, in which the stent can be accommodated with the heart valve prosthesis, can be made sufficiently small with respect to its external diameter. (77) The expandable heart valve stent with the heart valve prosthesis attached to it can be accommodated temporarily during implantation in the folded-up state in the catheter tip of the insertion system, which is provided at the distal end region of the catheter system. The catheter system may be of a length sufficient to allow the catheter tip provided at the distal end region of the catheter system to be guided through the aorta to the patient's heart by insertion at the patient's groin.

(78) The insertion system designed for transarterial or transfemoral access is therefore suitable for inserting a heart valve stent with a heart valve prosthesis attached to it, transarterially or transfemorally into the body of the patient; for example, the catheter system of the insertion system is inserted with the catheter tip located at the distal end of the catheter system via puncture of the *A. femoris communis* (inguinal artery).

(79) In particular, with the insertion system designed for transarterial or transfemoral access, the catheter system may be designed so that it is both kink-resistant and flexible such that a bending radius of up to 4 cm, and preferably up to 3 cm, can be realised, at least at the distal end region of the catheter system.

Description

- (1) Preferred embodiments will be described with reference to the appended drawings below.
- (2) Of these:
- (3) FIG. 1: an embodiment of an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a part-sectioned side elevation;
- (4) FIG. 2: an embodiment of a handle for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a part-sectioned side elevation;
- (5) FIG. 3a: an embodiment of an insertion system for transfemoral/transarterial insertion of a heart valve stent in a side elevation;
- (6) FIG. 3b: a side elevation of the transfemoral transarterial insertion system in accordance with FIG. 3a with a deflected catheter system;
- (7) FIG. 4: a further embodiment of an insertion system for transfemoral/transarterial insertion of a heart valve stent in a side elevation;
- (8) FIG. 5: a further embodiment of an insertion system for transfemoral/transarterial insertion of a heart valve stent in a side elevation;
- (9) FIG. 6a-d: side elevations of the transfemoral transarterial insertion system in accordance with FIG. 3a in its four previously defined functional states to illustrate the loading procedure of the insertion system
- (10) FIGS. 7a-d: side elevations of the transfemoral transarterial insertion system in accordance with FIG. 3a in its four previously defined functional states to illustrate the release procedure of a stent housed in the catheter tip of the insertion system;
- (11) FIG. 8: an embodiment of a catheter tip for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a part-sectioned side elevation;
- (12) FIG. 9: a further embodiment of a catheter tip for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a part-sectioned side elevation;
- (13) FIG. 10a: an exemplary embodiment of a catheter shaft for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a cross-sectional elevation;

(14) FIG. **10b**: a further exemplary embodiment of a catheter shaft for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in a cross-sectional elevation;

(15) FIG. **11**: a schematic view to illustrate a transfemoral/transarterial implantation procedure of a heart valve stent;

(16) FIG. **12a-c**: three-dimensional schematic part-sectioned view of the catheter tip of a transfemoral/trans-apical insertion system in different functional states to illustrate the implantation procedure of a heart valve stent mounted in the catheter tip;

(17) FIG. **13a-d**: side elevations of a further embodiment of a catheter tip for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent in its four previously defined functional states to illustrate the release procedure of a stent housed in the catheter tip of the insertion system; and

(18) FIG. **13e**: a side elevation of the embodiment of a catheter tip in accordance with FIG. **13a-d** in its state after releasing a stent housed in the catheter tip and ready to be removed again from the body of the patient.

(19) FIG. **11** shows schematically an example of how a transarterial or transfemoral access can be gained to the heart of a patient. In the illustration in accordance with FIG. **11**, a heart valve stent **150** is advanced with the aid of a insertion system **100** via the femoral artery to the aortic valve. Embodiments of an insertion system **100**, which is suitable for transarterial or transfemoral access, are described in the following.

(20) In accordance with a preferred embodiment, an insertion system **100** has a catheter system **1** and a handle **70** connected to the proximal end section of the catheter system **1**. As depicted, for example, in FIG. **1**, the catheter system **1** of the preferred embodiment comprises a catheter tip **10** having a seat portion for accommodating a stent to be inserted in its collapsed state and a stent holder **15** for releasably fixing the stent. The catheter system **1** further comprises a catheter shaft **30** for connecting the catheter tip **10** to the handle **70** of the insertion system **100**, the distal end section of the catheter shaft **30** being flexible enough such that the catheter tip **10** and the distal end section of the catheter shaft **30** may pass the aortic arch during insertion through the aorta of the patient.

(21) The seat portion of the catheter tip **10** comprises a first sleeve-shaped member **11** and a second sleeve-shaped member **21**, the cross-section of the second sleeve-shaped member **21** are preferably identical to each other such that the first and second sleeve-shaped member **11**, **21** can completely enclosed a stent accommodated in the catheter tip **10**. In addition, the first and second sleeve-shaped members **11**, **21** are movable relative to each other and relative to the stent holder **15**.

(22) For this purpose, first force transmitting means **31** with a distal end section connected to the first sleeve-shaped member **11** and a proximal end section connected to first operating means **71** of the handle **70** are provided. In addition, second force transmitting means **41** with a distal end section connected to the second sleeve-shaped member **21** and a proximal end section connected to second operating means **81** of the handle **70** are provided. When manipulating the first and/or second operating means **71**, **81** of the handle **70**, the first and/or second sleeve-shaped members **11**, **21** may be moved relative to each other and relative to the stent holder **15**.

(23) As can be seen from FIG. **10a** and FIG. **10b**, the first force transmitting means **31** may be constituted by a first catheter tube **32** defining a first lumen and the second force transmitting means **41** is constituted by a second catheter tube **42** defining a second lumen. The second catheter tube **42** may have a cross-section less than the cross-section of the first catheter tube **32**. The first catheter tube **32** may be disposed concentrically and coaxially with the second catheter tube **42** and the second catheter tube **42** is received within the first lumen defined by the first catheter tube **32**.

(24) Contrary to the first and second sleeve-shaped members **11**, **21** of the catheter tip **10**, however, the stent holder **15** of the catheter tip **10** is not moveable relative to the handle **70** of the insertion system **100**. Rather, the stent holder **15** is connected to the housing **70'** of the handle **70** by using a stent holder tube **62** having a distal end connected to the stent holder **15** and a proximal end connected to a body **70'** of the handle **70**.

(25) Referring to FIG. 10b, the stent holder tube 62 may have a cross-section less than the cross-section of the first catheter tube 32. In particular, the first catheter tube 32 may be disposed concentrically and coaxially with both, the second catheter tube 42 on the one hand and the stent holder tube 62 on the other hand. Preferably, the stent holder tube 62 has a cross-section less than the cross-section of the first catheter tube 32 and greater than the cross-section of the second catheter tube 42 such that the stent holder tube 62 is received within the first lumen defined by the first catheter tube 32 and the second catheter tube 42 is received within a passageway defined by the stent holder tube 62. The passageway defined by the stent holder tube 62 has a diameter sufficient to accommodate the second catheter tube 42 such that the second catheter tube 42 is moveable relative to the stent holder tube 62.

(26) The second lumen defined by the second catheter tube 42 has a diameter sufficient to accommodate a guide wire 180. The second catheter tube 42 may be made from a rigid material including, for example, nitinol, stainless steel or a rigid plastic material (see FIG. 10b). The material of the distal end section of the second catheter tube 42 may have an increased flexibility compared to the material of the proximal end section in order to allow the distal end section of the catheter shaft 30 to pass the aortic arch during insertion of the catheter tip 10. For example, the guiding tube 52 may be a 17F-catheter tube and the first catheter tube 32 may be a 12F-catheter tube.

(27) As can be seen, for example, from FIG. 9, the distal end section of the second catheter tube 42 terminates in a soft catheter end tip 25 having an atraumatic shape. The soft catheter end tip 25 is provided with a channel aligned with the second lumen defined by the second catheter tube 42 such that a guide wire 180 accommodated within the second lumen of the second catheter tube 42 may pass through the channel of the soft catheter end tip 25. The second sleeve-shaped member 21 of the catheter tip 10 is connected to the soft catheter end tip 25 such that the opened end of the second sleeve-shaped member 21 faces in the proximal direction opposite to the direction of the soft catheter end tip 25 and to the second catheter tube 42.

(28) According to the exemplary embodiment depicted in FIG. 10b, the stent holder tube 62 is made of a rigid material, for example, a rigid plastic material, stainless steel or nitinol. The distal end of the stent holder tube 62 terminates in the stent holder 15 which is also made of a rigid material, for example, a rigid plastic material or stainless steel. The passageway defined by the stent holder tube 62 is aligned with a channel which passes through the stent holder 15. In this way, the second catheter tube 42 is accommodated in the passageway of the stent holder tube 62 and the channel of the stent holder 15 such as to be moveable relative to the stent holder tube 62 and the stent holder 15.

(29) The first catheter tube 32 is made of a bendable but inelastic material. For example, the first catheter tube 32 may be at least partly made of a braided or non-braided catheter tube. The first catheter tube 32 shall be adapted to transfer compression and tension forces from the first operating means 71 of the handle 70 to the first sleeve-shaped member 11 of the catheter tip 10 without overly changing its total length. The distal end of the first catheter tube 32 terminates at a flared section as a transition to the section defining the first sleeve-shaped member 11 of the catheter tip 10.

(30) As can be seen from FIG. 9, the flared section and the first sleeve-shaped member 11 may be formed integrally and may be connected to the distal end section of the first catheter tube 31. In addition, the flared section may constitute the first sleeve-shaped member 11 of the catheter tip 10. The first sleeve-shaped member 11 and the flared section of the first catheter tube 31 may be all of the same material and originating from the same raw tube prior to a widening process so that the flared section and the first sleeve-shaped member 11 are the same elements.

(31) Referring for example to FIG. 1, the insertion system 100 according to the preferred embodiment further comprises a guiding tube 52 having a cross-section greater than the cross-section of the first catheter tube 32. The guiding tube 52 defines a passageway and is disposed

concentrically and coaxially with the first catheter tube **32**, the stent holder tube **62** and the second catheter tube **42** such that the first catheter tube **32** with the stent holder tube **62** and the second catheter tube **42** accommodated therein is at least partly accommodated within the passageway defined by the guiding tube **52**, wherein the first catheter tube **32** is moveable relative to the guiding tube **52**. In particular, the guiding tube **52** terminates proximal to the catheter tip **10** wherein the cross-section of proximal end section of the guiding tube **52** shall be the same as or less than the cross-section of the flared section provided at the proximal end of the first catheter tube **32** so that a smooth transition from the first sleeve-shaped member **11** of the catheter tip **10** to the guiding tube **52** may be achieved (see FIG. **9**).

(32) The proximal end section of the guiding tube **52** terminates distal to the handle **70**. The proximal end section of the guiding tube **52** may be detached disconnected from the handle **70** so that the handle **70** as well as the first and second catheter tubes **32**, **42** and the stent holder tube **62** together with catheter tip **10** may be moved relative to the guiding tube **52**.

(33) The distal end of the guiding tube **52** is formed such that the flared section provided at the distal end section of the first catheter tube **32** may abut on the distal end of the guiding tube **52** without abrupt transition. The guiding tube **52** may be of a thin material such as to allow length deformation of the guiding tube **52** upon transfer of compression and tension forces. The material of the guiding tube **52**, however, shall have sufficient stiffness in order to mechanically avoid kinking of the flexible sections of the distal portion of the catheter shaft **30** during insertion of the catheter tip **10**.

(34) The proximal end of the guiding tube **52** is releasably connectable to the body **70'** of the handle **70**. In this way, the guiding tube **52** may have a double-function:

(35) In case, the proximal end of the guiding tube **52** is connected to the handle **70**, the guiding tube **52** serves as a distal extension of the body **70'** of the handle **70** relative to which the first and second operating means **71**, **81** are moveable for manipulating the first and second sleeve-shaped members **11**, **21** of the catheter tip **10**. Hence, the position of the stent holder **15** relative to the native heart valve of the patient may be changed by moving the guiding tube **52** connected to the handle **70**.

(36) In case, the proximal end of the guiding tube **52** is not connected to the body **70'** of the handle **70**, the guiding tube **52** may serve as an introducer tube, i.e. as a portal for passing the catheter tip **10** of the catheter system **1** into the patient's body and up to the heart.

(37) As depicted, for example, in FIG. **1**, an inlet port **53** may be provided at a proximal end section of the guiding tube **52** for injection of fluids into the guiding tube **52**. Furthermore, a check valve may be provided at the proximal end section of the guiding tube **52** to prevent fluid from leaking out of the guiding tube **52**.

(38) A description is given in the following, with reference to FIGS. **1** to **10b**, of the components of exemplary embodiments of insertion systems **100**, which are suitable for a transarterial or transfemoral access to the implantation location. During a transarterial or transfemoral access, the catheter tip **10** of the insertion system **100** is advanced, for example, via the aorta to the implantation site.

(39) FIG. **1** shows a part-sectioned representation of an exemplary embodiment of an insertion system **100** designed for transfemoral or transarterial access.

(40) As illustrated in FIG. **1**, an insertion system **100** according to the present disclosure may comprise a catheter system **1** and a handle **70** connected to the proximal end section of the catheter system **1**. The catheter system **1** comprises a catheter tip **10** and a catheter shaft **30** for connecting the catheter tip **10** to the handle **70**. The catheter tip **10** has a seat portion for accommodating a stent (see FIGS. **12a-c**) in its collapsed state as well as a stent holder **15** for releasably fixing the stent.

(41) The seat portion of the catheter tip **10** is constituted by a first sleeve-shaped member **11** and a second sleeve-shaped member **21**. As will be explained in more detail with reference to FIGS. **6a-d**

and FIGS. 7a-d, the sleeve-shaped members **11**, **21** of the catheter tip **10** are movable relative to each other and relative to the stent holder **15**.

(42) The catheter shaft **30** comprises first force transmitting means **31**, second force transmitting means **41** and guiding means **51**. In accordance with the exemplary embodiment depicted in FIG. **1**, the first and second force transmitting means **41** **31**, **41** of the catheter system **1** are realized as flexible, elongated catheter tubes **32**, **42**. Each of the first and second catheter tubes **32**, **42** defines a separate lumen. In addition, the guiding means **51** is realized as guiding tube **52** defining a passageway within which the first and second catheter tubes **32**, **42** are received such as to be movable relative to the guiding tube **52**.

(43) As can be seen in FIG. **1**, the guiding tube **52** has a distal end which terminates proximal to the catheter tip **10**. On the other hand, the first catheter tube **32** has a length which is the same as, or substantially similar to the length of the second catheter tube **42**. The first catheter tube **32** terminates at its distal end in a flared section as a transition to the section with wider cross-section defining the first sleeve-shaped member **11** of the catheter tip **10**. In particular, and as can be seen from the illustration in FIG. **9**, the wider section of the first catheter tube **32** is formed integrally with the distal end section of the first catheter tube **32**. The wider section has a length greater than the length of a collapsed stent to be accommodated in the catheter tip **10**.

(44) As already mentioned, in the exemplary embodiment depicted in FIG. **1**, the first force transmitting means **31** of the catheter system **1** is constituted by a first catheter tube **32** defining a first lumen, wherein the second force transmitting means **41** is constituted by a second catheter tube **42** defining a second lumen. The second catheter tube **42** has a cross-section less than the cross-section of the first catheter tube **32**, wherein the first catheter tube **32** is disposed concentrically and coaxially with the second catheter tube **42**. The cross-section of the catheter tip **10**, however, is greater than or equal to the cross-section of the guiding tube **52**.

(45) On the other hand, the guiding tube **52** has a cross-section which is greater than the cross-section of the part of the first catheter tube **32** which is received within the guiding tube **52**. The cross-section of the catheter tip **10**, however, is greater than the cross-section of the guiding tube **52**. Hence, the guiding tube **52** cannot be removed from the insertion system **100** without disconnecting the catheter system **1** from the handle **70**.

(46) At the proximal end section of the guiding tube **52**, a check valve may be provided for preventing fluid from leaking out of the guiding tube **52**. Furthermore, an inlet port **53** may be provided at the proximal end section of the guiding tube **52** for injection of fluids into the guiding tube **52**. Hence, fluids such as saline solution may be injected through the inlet port **52** to flush the interior passageway of the guiding tube **52** and to reduce the incidence of blood clotting. A stopcock may be attached to the inlet port **53** to maintain the port **53** in a closed position when the port **53** is not being accessed to flush the passageway of the guiding tube **52**.

(47) The guiding tube **52** is movable relative to the handle **70** and the first and second catheter tubes **32**, **42**. This provides a grip for the user who can hold the catheter shaft **30** at its proximal end section during positioning of the catheter tip **10** and during manipulation of the sleeve-shaped element **11** of the catheter tip **10**. The user can hold the guiding tube **52**, and in particular the proximal end section of the guiding tube **52** for supporting the movement of the first sleeve-shaped element **11** of the catheter tip **10** relative to the handle **70** such that the outer sheath of the catheter system **1** need not be held by the user or kinked.

(48) In the exemplary embodiment of the insertion system **100** depicted in FIG. **1**, a handle **70** is utilized, said handle **70** comprising first and a second operating means **71**, **81**, which are connected by means of corresponding first and second force transmission means **31**, **41** of the catheter shaft **30** to the first and second sleeve-shaped member **21**s **11**, **21** of the catheter tip **10**. The first operating means **71** has a first pusher **73** which is functionally connected to the first slide **74**. The first slide **74** is guided in a first guide **72** in the longitudinal direction L of the handle **70**. The distal-side end of the first guide **72** defines the first stop **75** and the proximal-side end of the first guide **72**

the second stop **76**, which define the overall longitudinal displacement that can be effected with the first operating means **71**. A locking element **77'** may be positioned between the distal-side and the proximal-side end of the first guide **72**, which defines the additional stop **77**.

(49) The second operating means **81** of the handle **70** shown in FIG. **1** has a second pusher **83**, which is functionally connected to a second slide **84**. The second slide **84** is guided in a longitudinal guide (second guide **82**) between a first stop **85** and a second stop **86**. The second slide **84** is connected by means of the second force transmission means **41** with the second sleeve-shaped member **21** of the catheter tip **10**. On actuation of the second operating means **81**, the second slide **84** is moved in the longitudinal direction L of the handle **70** from the first stop **85** to the second stop **86**. This movement effects a longitudinal displacement of the second sleeve-shaped member **21** of the catheter tip **10** connected via the second force transmission means **41** with the second operating means **81**.

(50) To prevent an unintended displacement of the second slide **84**, the second operating means **81** is equipped with a securing element **89**, which may connect the second slide **84** with the body **70'** of the handle **70** when in use. A longitudinal displacement of the second slide **84** to the second stop **86** is possible following removal or deactivation of the securing element **89**.

(51) FIG. **2** shows a further embodiment of a handle **70** of an insertion system **100** designed for transfemoral or transarterial access in a part-sectioned side view. The construction and mode of operation of the first and second operating means **81** **71**, **81** of the embodiment of the handle **70** shown in FIG. **2** is comparable in structural and functional respects to the handle **70** as previously described with reference to FIG. **1**. Hence, elements in FIG. **2** that are generally similar to previously described elements have the same reference numbers compared with the reference numbers in FIG. **1** previously used for the similar elements.

(52) In distinction to the handle **70** described with reference to FIG. **1**, however, the handle **70** in accordance with FIG. **2** is provided with a third operating means **96** in the form of a wheel, by means of which a flexural link region **34** of the catheter shaft **30** can be controlled. It is important to note, however, that the catheter shaft **30** is only optionally provided with such flexural link region **34**. Rather, the material of the distal end section of the catheter shaft **30** may have an increased flexibility compared to the material of the proximal end section in order to allow the distal end section of the catheter shaft to pass **30** the aortic arch during insertion of the catheter tip.

(53) In the exemplary embodiment depicted in FIG. **2**, the third operating element **96** preferably has a detent device **100**, to allow a set deflection of the flexural link region **34** of the catheter shaft **30** to be fixed. For example, it is possible to provide a suitable catch mechanism on the hand wheel of the third operating means **96**, which cooperates with the body **70'** of the handle **70**. In particular, it is possible for the flexural link region **34** of the catheter shaft **30** to be connected to the third operating means **96** by way of a control wire **35** whereby, on an actuation of the third operating means **96** via the control wire **35** a tensile force is exerted on the flexural link region **34**, which produces the deflection of the flexural link region **34** (see FIG. **3b**).

(54) However it is also possible, of course, to choose another embodiment as the third operating means **96** for deflecting a flexural link region **34** of the catheter shaft **30**, in case the catheter shaft **30** is provided with such a flexural link region **34**.

(55) The handle **70** of the insertion system **100** designed for transarterial or transfemoral access may be provided with a pretensioning device, shown in FIG. **2**. With such a pretensioning device, a constant tensile force may be exerted via the second operating means **81** on the second sleeve-shaped member **21** of the catheter tip **10**. As shown in FIG. **2**, the pretensioning device may have a compression spring **97**, permanently stressed along its spring axis, which is prestressed between a first stop **97a** connected to the body **70'** of the handle **70** and a second stop **97b** connected to the proximal end region of the second operating means **81**. In this respect, a permanent, previously defined or definable tensile force is exerted on the second sleeve-shaped member **21** of the catheter tip **10**.

(56) The pretensioning device implemented with the spring **97** in the embodiment in accordance with FIG. **2** may be advantageous when the catheter shaft **30** is bent during the implantation procedure, for example, when the catheter tip **10** of the insertion system **100** is inserted through the aorta. When the catheter shaft **30** is bent, the outer fibres of the catheter shaft **30** are shortened. This can be compensated appropriately with the aid of the pretensioning device. In detail, on bending of the flexural link region **34** relative to the neutral fibres of the catheter shaft **30** running along the longitudinal axis **L**, the outer fibres of the catheter shaft **30** radially spaced from the neutral fibres are shortened. Since the second force transmission means **41**, which connects the second operating means **81** with the second sleeve-shaped member **21** in the insertion system **100**, normally runs along the neutral fibre of the catheter shaft **30**, a bending contraction inevitably occurs when the catheter shaft **30** is bent, having the result that, despite fixing of the first operating means **71**, the first sleeve-shaped member **11** of the catheter tip **10** is displaced relative to the stent holder **15** in a proximal direction.

(57) This longitudinal displacement of the first sleeve-shaped member **11** of the catheter tip **10** that takes place during the bending procedure is compensated with the aid of the prestressing device (spring **97**), since the spring **97** of the prestressing device exerts a constant tensile force on the second force transmission means **41** and therefore on the second sleeve-shaped member **21** of the catheter tip **10** and consequently constantly presses the distal-side end tip **25** of the catheter tip **10** against the distal-side end of the first sleeve-shaped member **11**. This enables the catheter tip **10** to remain completely closed even during a deflection of the catheter shaft **30** effected, for example, when the catheter tip **10** is inserted through the aorta.

(58) On actuation of the second operating means **81** of the handle **70**, it is necessary to press the second slide **84** against the prestress supplied by the spring **97** of the prestressing device on the second stop **86**.

(59) It is important to note, however, that a prestressing device of the kind as described above is not mandatory for the insertion system as disclosed herein.

(60) A further exemplary embodiment of an insertion system **100** designed for transarterial transfemoral access is shown in FIGS. **3a, b**. Elements in FIGS. **3a, b** that are generally similar to previously described elements have the same reference numbers compared with the reference numbers in FIGS. **1** and **2** previously used for the similar elements.

(61) The insertion system **100** shown in FIGS. **3a, b** comprises a catheter system **1** of the kind as previously described with reference to FIG. **1**, i.e. a catheter system **1** having a catheter tip **10** and a catheter shaft **30** which is provided with a first catheter tube **32** acting as first force transmitting means **31**, a second catheter tube **42** acting as second force transmitting means **41**, and a guiding tube **52** acting as guiding means **51**. Contrary to the catheter shaft **30** utilized in the exemplary embodiment of the insertion system **100** depicted in FIG. **1**, however, the catheter shaft **30** of the insertion system **100** shown in FIGS. **3a, b** is provided with a flexural link region **34** of the kind as previously described with reference to FIG. **2**.

(62) As will be described in the following, the insertion system **100** shown in FIGS. **3a, b** is provided with a different embodiment of a handle **70** which is used in order to manipulate the first and second sleeve-shaped members **11, 21** of the catheter tip **10**.

(63) In relation to the handle **70** used with the insertion system **100** shown in FIG. **3a**, it can be seen that the end region of the handle **70** is in the form of a turning mechanism **98** (rotatable means), with which the second force transmission means **41** of the catheter shaft **30** can be twisted with the distal-side end tip **25** and the second sleeve-shaped member **21** of the catheter tip **10** fastened to it about the longitudinal axis **L** of the catheter tip **10**. The second sleeve-shaped member **21** of the catheter tip **10** is connected by means of a loose bearing to the stent holder **15**, allowing transmission of a turning moment between the second sleeve-shaped member **21** and the stent holder **15**, without allowing transmission of any tensile or compression forces acting in the direction of the longitudinal axis **L** of the catheter tip **10**. Thus, when a turning movement of the

second sleeve-shaped member **21** is induced with the turning mechanism **98**, the stent holder **15** also turns correspondingly about the longitudinal axis L.

(64) The turning mechanism **98** preferably allows the stent holder **15** to twist through approximately 120°. Thus the rotation of a stent housed in the catheter tip **10**, and particularly the positioning hoops already released in the second functional state of the insertion system **100**, can be controlled, facilitating precise positioning of the already expanded positioning hoops of the stent in the pockets of the insufficient, native heart valve.

(65) Preferably, the rotation movement of the stent holder **15** about the longitudinal axis L of the catheter tip **10** that can be effected with the turning mechanism **98** exhibits a previously definable, preferably small delay in reaction to a turning moment initiated by means of the turning mechanism **98**.

(66) Further, the embodiment of the handle **70** shown in FIG. **3a** is equipped with a third operating means **96** in the form of a wheel, with which a flexure/link **34**, preferably provided at the distal end region of the catheter shaft **30**, can be deflected.

(67) The deflection of the distal end region of the catheter shaft **30** that can be effected with this flexural link region **34** is shown schematically in FIG. **3b**. In detail, a device is provided for force transmission (control wire **35**—see FIG. **8**) which is connected on one side to the flexural link regions **34** preferably provided at the distal end region of the catheter shaft **30** and, on the other side, to the third operating means **96** of the handle **70** implemented in the embodiment shown in FIG. **3** as a hand wheel.

(68) It is possible to implement the device for force transmission as a control wire **35**, which is passed through the inside of the first transmission means **31** and preferably at the distal end of the flexural link region **34** or at the proximal end of the catheter tip **10** (see FIG. **8**) to have a directed effect on the curvature of the flexural link region **34**. With the tensile forces that can be exerted on the flexural link region **34** with the aid of the control wire **35**, it is possible to obtain a defined curvature of the distal end region of the catheter shaft **30**. This is a particular advantage during transarterial/transfemoral access when navigating the aortic arch.

(69) Further exemplary embodiments of an insertion system **100** which is suitable for transarterial/transfemoral access to the implantation location are shown in FIGS. **4** and **5**. Elements in FIGS. **4** and **5** that are generally similar to previously described elements have the same reference numbers compared with the reference numbers in FIGS. **1**, **2** and **3a, b** previously used for the similar elements.

(70) Compared with the exemplary embodiment depicted in FIGS. **1** and **2** as well as FIGS. **3a, b**, the embodiments shown in FIGS. **4** and **5** differ first and foremost in relation to the implementation of the corresponding operating means **71, 81** of the handle **70**.

(71) The insertion system **100** in accordance with FIG. **4** has a handle **70** with which the first operating means **71**, which is used for manipulation of the first sleeve-shaped member **11** of the catheter tip **10**, is similar to a trigger of a revolver. The user such as a physician who carries out the treatment may hold the handle **70** at the grip **88**, while the first operating means **71** in the form of a trigger of a revolver is operated with the index finger of the hand holding it.

(72) In the insertion system **100** shown in FIG. **5**, a handle **70** is used which corresponds in structural and functional respects to the handle **70** used with the insertion system **100** in FIG. **3** with the exception of the grip **88** provided in the embodiment in accordance with FIG. **3**.

(73) A description is given in the following, with reference to FIGS. **6a-d** and FIGS. **7a-d**, of the functional coaction of the components of an insertion system **100**, which is suitable for a transarterial or transfemoral access to the implantation location. Elements in FIGS. **6a** to **6d** and FIGS. **7a** to **7d** that are generally similar to previously described elements have the same reference numbers compared with the reference numbers in FIGS. **1** to **5** previously used for the similar elements.

(74) Reference is made to FIGS. **6a** to **6d** for illustrating the procedure for loading a stent into the

catheter tip **10** of the insertion system **100** in FIGS. **7a** to **7d**, the stepwise release of a stent mounted in the catheter tip **10** of the insertion system **100** is illustrated.

(75) It is important to note, however, that the procedure for loading a stent into the catheter tip **10** as depicted in FIGS. **6a** to **6d**, as well as the procedure for stepwise releasing of a stent mounted in the catheter tip **10** as depicted in FIGS. **7a** to **7d** also apply to the other exemplary embodiments of the transarterial transfemoral insertion system **100** disclosed herein.

(76) The handle **70** for the transarterial J transfemoral insertion system **100** according to the illustration in FIGS. **6** and **7** has a wheel rotatably mounted in the handle **70** which is functionally connected to the first sleeve-shaped member **11** of the catheter tip **10** associated with the first operating means **71** via a corresponding first force transmission means **31**, so that force can be directly transmitted from the first operating means **71** in the form of the wheel to the first sleeve-shaped member **11** of the catheter tip **10**.

(77) In detail, it is provided that, with the first operating means **71** of the handle **70** in accordance with FIG. **6** and FIG. **7**, the first operating means **71** in the form of the wheel can turn between a first stop and a second stop, in order to execute a definable longitudinal displacement stroke on the first sleeve-shaped member **11** of the catheter tip **10**. The first operating means **71** of the handle **70** is provided with a additional stop between the first and second stop which cooperates, on one side with the first stop and on the other up with the second stop so that, on actuation of the first operating means **71**, a longitudinal displacement of the first sleeve-shaped member **11** of the catheter tip **10** can be effected relative to the stent holder **15** of the catheter tip **10**, consisting of two defined separate steps.

(78) With the first operating means **71** used in the form of a wheel, the additional stop associated with the first operating means **71** is in the form of a locking element **77'** positioned removably in the flow of force between the wheel and the first sleeve-shaped member **11** of the catheter tip **10**, interrupting direct force transmission from the wheel to the first sleeve-shaped member **11** of the catheter tip **10**. Alternatively, however, it is possible for the additional stop associated with the first operating means **71** to be in the form of a locking element restricting the free rotation of the wheel between the first and the second stop.

(79) However, it is of course also possible in principle for the first operating means **71** of the handle **70** used with the insertion system **100** designed for transarterial transfemoral access not to be a wheel, but to be implemented as a pusher mechanism.

(80) In relation to the handle **70** that is used with the embodiment of the insertion system **100**, for example in accordance with the illustrations in FIGS. **6** and **7**, it is provided that the second operating means **81** has a second slide **84** guided in a second guide **82** and functionally connected to a second pusher **83**. This second slide **84**, which is guided in the interior of the handle **70** and therefore cannot be seen in the view of FIGS. **6** and **7**, is functionally connected to the second sleeve-shaped member **21** of the catheter tip **10** associated with the second operating means **81** by means of a second force transmission means **41** so that, on actuation of the second operating means **81**, force is directly transmitted from the second slide **84** to the second sleeve-shaped member **21** of the catheter tip **10**.

(81) The second operating means **81** can be displaced between a first position (Pos. 1) and a second position (Pos. 2) in the longitudinal direction of the handle **70**, whereby the longitudinal displacement stroke that can be thus effected via the second force transmission means **41** is transferred directly to the second sleeve-shaped member **21** of the catheter tip **10**. The first and second positions are each defined with the aid of a first and a second stop **85**, **86**.

(82) A securing element **89** is provided, associated with the second operating means **81**, which is removably located on the second guide **82** and which blocks longitudinal displacement of the (second) slide **84** associated with the second operating means **81** when used.

(83) The handle **70** which is used with the transarterial transfemoral insertion system **100** of the embodiment shown in FIGS. **6** and **7** further exhibits an optional grip **88**, which facilitates the

operability of the handle **70** and in particular the operating conformity of the handle **70**. The grip **88** is preferably releasably connected to the body **70'** of the handle **70** and can optionally be fixed at different positions on the body **70'** of the handle **70**.

(84) In relation to the construction of the catheter tip **10** which is used, for example, with the insertion system **100** shown in FIGS. **6** and **7** and which allows transarterial/transfemoral access of a stent housed in the catheter tip **10** to the implantation location, it can be seen from FIGS. **6** and **7** that the catheter tip **10** has a stent holder **15** for releasably fixing of, for example, the second retaining region of a stent that can be housed in the catheter tip **10**. The retaining elements **16** of the stent holder **15** in the form of a crown are provided at the proximal end of the stent holder **15**.

(85) Further, the catheter tip **10** of the insertion system **100** designed for transarterial/transfemoral access comprises a mounting device for mounting a heart valve stent, where required, with a heart valve prosthesis fastened to it. In detail, the mounting device of the catheter tip **10** consists of a first sleeve-shaped member **11**, particularly for accommodating the positioning hoops of a stent, and a second sleeve-shaped member **21**, in particular for accommodating the heart valve prosthesis fastened to it, when required.

(86) The first operating means **71** of the handle **70** co-operates in the embodiment according to FIGS. **6** and **7** with the first sleeve-shaped member **11** of the catheter tip **10** so that, on actuation of the first operating means **71**, by transfer of a defined longitudinal displacement stroke, a previously definable longitudinal displacement of the first sleeve-shaped member **11** can be effected relative to the stent holder **15**. On the other hand, with the insertion system **100** according to FIGS. **6** and **7**, the second operating means **81** of the handle **70** co-operates with the second sleeve-shaped member **21** of the catheter tip **10** so that, on actuation of the second operating means **81**, by transfer of a defined longitudinal displacement stroke, a previously definable longitudinal displacement of the second sleeve-shaped member **21** of the catheter tip **10** relative to the stent holder **15** can be effected.

(87) The second sleeve-shaped member **21**, which is used to house the retaining hoops of the stent with, where required, the heart valve prosthesis fastened to them, is located at the distal end region of the catheter tip **10**, while the first sleeve-shaped member **11** is located between the second sleeve-shaped member **21** and the handle **70**.

(88) In the insertion system **100** shown in FIGS. **6** and **7**, the second force transmission means **41**, which connects the second operating means **81** of the handle **70** to the second sleeve-shaped member **21** of the catheter tip **10**, is preferably in the form of an inner catheter running inside the interior of the catheter or tube system. The first force transmission means **31**, which connects the first operating means **71** of the handle **70** to the first sleeve-shaped member **11** of the catheter tip **10**, is in the form of an outer catheter, in the interior of which the first force transmission means **31** runs in the form of the inner catheter.

(89) On actuation on the second operating means **81**, the second sleeve-shaped member **21** can be moved relative to the stent holder **15** in the longitudinal direction **L** of the catheter tip **10** in a distal direction, thus away from the handle **70**, while, on actuation of the first operating means **71** of the handle **70**, the first sleeve-shaped member **11** of the catheter tip **10** can be moved relative to the stent holder **15** in the longitudinal direction **L** of the catheter tip **10** in a proximal direction, and thus towards the handle **70**.

(90) The manipulations of the respective sleeve-shaped members **11**, **21** of the catheter tip **10** that can be effected on actuation of the respective operating means **71**, **81** with the insertion system **100** of **100** designed for transarterial/transfemoral access in accordance with FIGS. **6** and **7** are described in detail in the following, with reference in particular to FIGS. **7a** to **7d**.

(91) An embodiment of a transarterial/transfemoral insertion system **100** is shown in its four different functional states in FIGS. **7a** to **7d**. In detail, the insertion system **100** is shown in its first functional state in FIG. **7a**, in which the catheter shaft **30** with the catheter tip **10** and, where required, with the stent accommodated in it can be inserted into the patient transarterially or

transfemorally and advanced via the aorta to the implantation site.

(92) In the first functional state of the insertion system **100** in accordance with FIG. **7a**, the catheter tip **10** is completely closed, whereby the two sleeve-shaped members **11**, **21** of the catheter tip **10** overlap telescopically. The respective diameters of the sleeve-shaped members **11**, **21** are chosen so that the folded-up retaining hoops of a stent, with the heart valve prosthesis fastened to them where required, can be housed in the second sleeve-shaped member **21**. The folded-up positioning hoops of the stent housed between the second sleeve-shaped member **21** and the first sleeve-shaped member **11** are held together in their folded form.

(93) The second retaining region of the stent is shown in the first functional state of the insertion system **100**, as shown in FIG. **7a**, with the stent holder **15** fixed at the proximal end of the catheter tip **10**. For this purpose, the retaining elements (retaining rings etc.) provided at the second retaining region of the stent are engaged with retaining elements **16** of the stent holder **15**.

(94) The retaining elements **16** of the stent holder **15** are covered by the first sleeve-shaped member **11** of the catheter tip **10** in the first functional state shown in FIG. **7a**, so that an engagement between retaining elements provided on the second retaining region of a stent and retaining elements **16** of the stent holder **15** would be possible.

(95) The first functional state of the insertion system **100** shown in FIG. **1a** is maintained during the transarterial insertion procedure. On reaching the implantation location, the insertion system **100** is transferred from the first functional state shown in FIG. **7a** to the second functional state shown in FIG. **7b**, by transferring the first operating means **71** (shown in the embodiment of the wheel in FIG. **7**) from the first position into the second position. The longitudinal displacement stroke transferred by actuation of the first operating means **71** to the first sleeve-shaped member **11** of the catheter tip **10** effects a displacement of the first sleeve-shaped member **11** relative to the stent holder **15** in the proximal direction, thus towards the handle **70**.

(96) The longitudinal displacement stroke executed on the first sleeve-shaped member **11** of the catheter tip **10** during the transition from the first functional state (see FIG. **7a**) to the second functional state (see FIG. **7b**) by the first operating means **71** of the handle **70** via a corresponding first force transmission means **31** is previously defined so that the first sleeve-shaped member **11** is displaced relative to the stent holder **15** in the proximal direction just so far that the positioning hoops of a stent housed in the catheter tip **10** would be released, though the distal end of the first sleeve-shaped member **11** of the catheter tip **10** would still cover the retaining elements **16** of the stent holder **15**, so that the engagement between the retaining elements provided at the second retaining region of the stent and the retaining elements **16** of the stent holder **15** would be secure.

(97) Since the second sleeve-shaped member **21** is not manipulated during the transition from the first functional state into the second functional state, the first retaining region of a stent housed in the catheter tip **10** with the heart valve prosthesis fastened to it would continue to be housed in its folded together state in the sleeve-shaped element of the second sleeve-shaped member **21**.

(98) The positioning hoops of a stent housed in the catheter tip **10** released in the second functional state of the insertion system **100** are opened as a result of the radial forces acting on them and can thus be positioned in the pockets of the insufficient native heart valve. Following appropriate positioning of the positioning hoops of the stent in the pockets of the native heart valve, the insertion system **100** is transferred from the second functional state shown in FIG. **7b** into the third functional state shown in FIG. **7c**. This is done by manipulation of the second operating means **81**, after the securing element **89** associated with the second operating means **81** has been removed.

(99) On actuation of the second operating means **81**, the second sleeve-shaped member **21** of the catheter tip **10** associated with the second operating means **81** is moved relative to the stent holder **15** by a previously established longitudinal displacement stroke defined with the second operating means **81** in a distal direction, thus away from the handle **70**. The longitudinal displacement stroke acting on the second sleeve-shaped member **21** is chosen so that the sleeve-shaped member **21** no longer covers the first retaining region of a stent housed in the catheter tip **10** with the heart valve

prosthesis fastened to it, where required, and thus releases the first retaining region of the stent. Due to the action of the radial forces, the distal retaining region of the stent with the heart valve prosthesis attached to it, where required, unfolds completely.

(100) Since the first operating means **71** of the handle **70** and the associated first sleeve-shaped member **11** of the catheter tip **10** are not manipulated during the transition from the second functional state in accordance with FIG. **7b** into the third functional state in accordance with FIG. **7c**, the distal end region of the first sleeve-shaped member **11** continues to cover the retaining elements **16** of the stent holder **15**, so that the engagement between the retaining elements of a stent housed in the catheter tip **10** and the retaining elements **16** of the stent holder **15** is secure and the proximal retaining region of the stent is in its folded-up state. This anchorage of the stent to the catheter tip **10** of the insertion system **100** allows an explantation of a stent that is already partially unfolded by returning the insertion system **100** from the third functional state, by appropriate manipulation of the second operating means **81** of the handle **70**, to the second functional state and then by suitable actuation of the first operating means **71** transfer to the first functional state.

(101) If an explantation of the stent with the heart valve prosthesis attached to it, where required, is unnecessary, the insertion system **100** is transferred from the third functional state shown in FIG. **7c** into the fourth functional state shown in FIG. **7d**, by turning the first operating means **71** of the handle **70** further from the second position to the third position after removal of the securing element **79** (locking element). This manipulation of the first operating means **71** that can be effected after removal of the securing element **79** results in a further defined movement of the first sleeve-shaped member **11** relative to the stent holder **15** of the catheter tip **10** in a proximal direction, thus towards the handle **70**. The longitudinal displacement stroke executed on the first sleeve-shaped member **11** is chosen so that the distal end of the first sleeve-shaped member **11** no longer covers the retaining elements **16** of the stent holder **15**, as a result of which an engagement between the retaining elements of a stent housed in the catheter tip **10** and the retaining elements **16** of the stent holder **15** can be released, which would also lead to a complete release of the second retaining region of the stent and a complete separation of the stent from the catheter tip **10** and correspondingly to a complete unfolding of the stent.

(102) The four functional states of the insertion system **100** designed for transarterial/transfemoral access, previously described with reference to FIGS. **7a** to **7d**, are shown in reverse order in FIGS. **6a** to **6d** to clarify the procedure for loading a stent into the catheter tip **10** of the insertion system **100**. Comparison between FIGS. **6a** to **6d** and FIGS. **7a** to **7d** show that the insertion system **100** can be loaded with a heart valve stent by transferring the insertion system **100**, starting from its fourth functional state in accordance with FIG. **6a** (see FIG. **7d**), into its third functional state in accordance with FIG. **6b** (see FIG. **7c**) after a stent has been positioned between the stent holder **15** on the second sleeve-shaped member **21** with its first retaining region in the direction of the second sleeve-shaped member **21**. Then the remaining functional states of the insertion system **100** are taken up in steps until the insertion system **100** shown in FIG. **6d** is finally in its first functional state with the closed catheter tip **10**.

(103) Reference is made to FIG. **9** for describing an exemplary embodiment of the catheter tip **10**. Elements in FIG. **9** that are generally similar to previously described elements have the same reference numbers compared with the reference numbers in FIGS. **1** to **7** previously used for the similar elements.

(104) An exemplary embodiment of a catheter shaft **30** is described in the following, with reference to the illustration in FIG. **9**. This catheter shaft **30** can be used with an insertion system **100** designed for transarterial or transfemoral access.

(105) In detail, FIG. **9** shows an exemplary embodiment of a shaft for an insertion system **100** in a cross-sectional elevation.

(106) The catheter shaft **30** exhibits a first force transmission means **31** in the form of a first catheter tube **32**, whereby this first catheter tube **32** is used to connect the first operating means **71**

of the handle **70** to the first sleeve-shaped member **11** of the catheter tip **10**. As can be seen in particular from the illustration in FIG. **1**, the first force transmission means **31** implemented as a first catheter tube **32** may be clamped between a screw cap **74'** and the first slide **74** of the first operating means **71** and consequently is permanently connected to the first slide **74**. The distal-side end region of the first catheter tube **32** merges into the first sleeve-shaped member **11** of the catheter tip **10** in the form of the stent sheath.

(107) The second force transmission means **41** of the catheter shaft **30** used with an insertion system **100** designed for transarterial or transfemoral access is preferably implemented as a second catheter tube **42**. The proximal-side end region of the second catheter tube **42** is connected to the second operating means **81** of the handle **70**. The distal-side end region of the second catheter tube **42** is connected to the catheter end tip **25** of the catheter tip **10**. The second sleeve-shaped member **21** of the catheter tip **10** is permanently connected by means of its distal-side end to the end tip **25** of the catheter tip **10** so that, on actuation of the second operating means **81** via the force transmission means **41** in the form of the second catheter tube **42**, a tensile or compressive force can be transmitted to the second sleeve-shaped member **21** of the catheter tip **10**.

(108) The exemplary embodiment of the catheter tip **10** further comprises a stent holder **15** at the proximal end section of the catheter tip **10**. The stent holder **15** has a passageway extending there through. The distal end section of the second force transmitting means **41** (second catheter tube **42**) passes through the passageway of the stent holder **15** and terminates at the second sleeve-shaped member **21**.

(109) The respective sleeve-shaped members **11**, **21** of the catheter tip **10** can be manipulated by corresponding operating means **71**, **81** of a handle **70** (not shown in FIG. **9**). In detail, the first sleeve-shaped member **11** of the catheter tip **10** is connected with a first operating means **71** of a handle **70** by using a first force transmitting means **31**. On the other hand, the second sleeve-shaped member **21** of the catheter tip **10** is connected to a second operating means **81** of the handle **70** by using a second force transmitting means **41**. In a preferred embodiment of the catheter shaft **30**, the first force transmitting means **31** is constituted by a first catheter tube **32** defining a first lumen, wherein the second force transmitting means **41** is constituted by a second catheter tube **42** defining a second lumen. The second catheter tube **42** has a cross-section less than the cross-section of the first catheter tube **32**, wherein the first catheter tube **32** is disposed concentrically and coaxially with the second catheter tube **42**.

(110) As shown in FIG. **9**, the distal end section of the second catheter tube **42** passes through the opened front face of the second sleeve-shaped member **21** and terminates in a cone-shaped end tip **25** of the catheter system **1**, wherein the base of this cone-shaped end tip **25** defines the distal front face of the second sleeve-shaped member **21**.

(111) The end tip **25** of the catheter system **1** is preferably a soft catheter end tip, for example a soft polymeric catheter end tip.

(112) At its distal end, the first catheter tube **32** terminates after an intermediate flared section in a section with wider cross-section defining the first sleeve-shaped member **11** of the catheter tip **10**. As can be seen from FIG. **9**, the flared section is formed integrally with the distal end section of the first catheter tube **32**. The flared section has a length greater than the length of a collapsed stent to be accommodated in the catheter tip **10**, wherein the difference in the length between the flared section and the stent in its collapsed state represents the length of the stent holder **15**.

(113) The catheter shaft **30**, which is connected to the catheter tip **10** depicted in FIG. **9**, also comprises a guiding tube **52** of the kind as previously described with reference to the exemplary embodiment depicted in FIG. **1**.

(114) The distal end of the guiding tube **52** terminates proximal to the catheter tip **10**. The guiding tube **52** defines a passageway within which the first and second catheter tube **32**, **42** are received such as to be movable relative to the guiding tube **52**.

(115) The distal end of the guiding tube **52** may be tapered such that it abuts the first catheter tube

32 in one of its possible positions on the catheter shaft **30**.

(116) Reference is made to FIG. **10a**, which is a cross-sectional view of a catheter shaft **30** according to an exemplary embodiment.

(117) As can be seen from the illustration in FIG. **10a**, the second force transmission means **41** in the form of the second catheter tube **42** runs along the neutral fibre of the catheter shaft **30** inside the first catheter tube **32**. The space between the first catheter tube **32** and the second catheter tube **42** may be filled with a filler material, so that a filler body **40** is formed. The filler material is preferably a relatively elastic plastic material to allow the catheter shaft **30** to bend overall. The filler body **40** is used for connecting the stent holder **15** of the catheter tip **10** to the body **70'** of the handle **70**.

(118) Alternatively, a stent holder tube **62** may be used for connecting the stent holder **15** of the catheter tip **10** to the body **70** of the handle **70**. The stent holder tube **62** may have a distal end connected to the stent holder **15**, a proximal end connected to the body **70'** of the handle **70** and a passageway extending through the stent holder tube **62**. Preferably, the stent holder tube **62** has a cross-section less than the cross-section of the first catheter tube **32** and greater than the cross-section of the second catheter tube **42**, wherein the first catheter tube **32** is disposed concentrically and coaxially with the stent holder tube **62** thereby accommodating the stent holder tube **52** such that the first catheter tube **32** is moveable relative to the stent holder tube **62**. The passageway of the stent holder tube **62** shall have a diameter sufficient to accommodate the second catheter tube **42** such that the second catheter tube **42** is moveable relative to the stent holder tube **62**.

(119) As depicted in FIG. **1**, the filler body **40** (or the stent holder tube **62**) may be connected by means of a fixing **87** to the body **70'** of the handle **70**. The proximal-side end region of the stent holder **15** attaches at the distal-side end region of the filler body **40** (see FIG. **8**). The connection between the stent holder **15** and the filler body **40** is preferably chosen so that it allows rotation of the stent holder **15** relative to the filler body **40**. This is especially necessary for control of the rotation of the positioning hoops of the already partially released stent during the implantation procedure (see FIG. **12a**).

(120) As an alternative, the complete catheter system **1** can be rotated for appropriate positioning of a stent connected with the catheter tip **10** and, in particular the positioning hoops of an already partially released stent during the implantation procedure. This is possible due to an appropriate transmission of torque and the flexibility of the catheter system **1**.

(121) In case, a stent holder tube **62** is used for connecting the stent holder **15** of the catheter tip **10** to the body **70'** of the handle **70**, the stent holder tube **62** may be rotatable relative to the first and second catheter tubes **32**, **42** about the longitudinal axis **L** of the catheter system **1**. This will be described later in more detail with reference to the exemplary embodiment depicted in FIG. **10b**.

(122) On the other hand, the second force transmission means **41** in the form of the second catheter tube **42** can be turned about the longitudinal direction **L**, for example, by means of a rotatable cap **98** which may be provided at the proximal end region of the handle **70**. This rotary movement is transferred from the second catheter tube **42** direct to the end tip **25** of the catheter tip **10** and thus to the second sleeve-shaped member **21** of the catheter tip **10**.

(123) It is particularly preferred that the second catheter tube **42** runs through the body of the stent holder **15** and cooperates with the stent holder **15** with the aid of a suitable toothing, to transmit a turning moment exerted by means of the rotary cap of the handle **70** on the second catheter tube **42** to the stent holder **15**, while tensile or compression forces acting in the longitudinal direction **L** of the catheter tip **10** are not transmitted from the second catheter tube **42** to the stent holder **15**.

(124) As can also be seen in the illustration in FIG. **10a**, a least one fluid channel **43** may be provided in the filler body **40** of the catheter shaft **30**, connected at its proximal-side end to an injection adapter **99b** (see FIG. **2**) and at its distal-side end correspondingly to the catheter tip **10**, consequently ensuring supply of fluid to the catheter tip **10** and draining of fluid from the catheter tip **10**.

(125) Furthermore, a channel may be provided in the filler body **40** for accommodating a control wire (control wire **35**—see FIG. **8**), with an operating means may cooperate with a flexural link region, in case the catheter shaft **30** is provided with such a flexural link region (see FIG. **3** and FIG. **2**). In the illustration in FIG. **8**, the distal-side end of a control wire **35** is fixed to the proximal-side end region of the stent holder **15**.

(126) Reference is made to FIG. **10b**, which is a cross-sectional view of a catheter shaft **30** according to an alternative exemplary embodiment.

(127) According to the embodiment depicted in FIG. **10b**, the first force transmitting means **31** may be constituted by a first catheter tube **32** defining a first lumen and the second force transmitting means **41** is constituted by a second catheter tube **42** defining a second lumen. The second catheter tube **42** may have a cross-section less than the cross-section of the first catheter tube **32**. The first catheter tube **32** may be disposed concentrically and coaxially with the second catheter tube **42** and the second catheter tube **42** is received within the first lumen defined by the first catheter tube **32**.

(128) A stent holder tube **62** is provided for connecting the stent holder **15** to the handle **70**, said stent holder tube **62** having a distal end connected to the stent holder **15** and a proximal end connected to a body **70'** of the handle **70**.

(129) As can be seen from FIG. **10b**, the stent holder tube **62** may have a cross-section less than the cross-section of the first catheter tube **32**. In particular, the first catheter tube **32** may be disposed concentrically and coaxially with both, the second catheter tube **42** on the one hand and the stent holder tube **62** on the other hand. Preferably, the stent holder tube **62** has a cross-section less than the cross-section of the first catheter tube **32** and greater than the cross-section of the second catheter tube **42** such that the stent holder tube **62** is received within the first lumen defined by the first catheter tube **32** and the second catheter tube **42** is received within a passageway defined by the stent holder tube **62**. The passageway defined by the stent holder tube **62** has a diameter sufficient to accommodate the second catheter tube **42** such that the second catheter tube **42** is moveable relative to the stent holder tube **62**.

(130) The second lumen defined by the second catheter tube **42** has a diameter sufficient to accommodate a guide wire **180**. The second catheter tube **42** may be made from a rigid material including, for example, nitinol, stainless steel or a rigid plastic material. The material of the distal end section of the second catheter tube **42** may have an increased flexibility compared to the material of the proximal end section in order to allow the distal end section of the catheter shaft **30** to pass the aortic arch during insertion of the catheter tip **10**. For example, the guiding tube **52** may be a 17F-catheter tube and the first catheter tube **32** may be a 12F-catheter tube.

(131) According to the exemplary embodiment depicted in FIG. **10b**, the stent holder tube **62** is made of a rigid material, for example, a rigid plastic material, stainless steel or nitinol. The distal end of the stent holder tube **62** terminates in the stent holder **15** which is also made of a rigid material, for example, a rigid plastic material or stainless steel. The passageway defined by the stent holder tube **62** is aligned with a channel which passes through the stent holder **15**. In this way, the second catheter tube **42** is accommodated in the passageway of the stent holder tube **62** and the channel of the stent holder **15** such as to be moveable relative to the stent holder tube **62** and the stent holder **15**.

(132) The embodiments of the insertion system **100** designed for transarterial/transfemoral access may have a first injection adapter **99a** at the proximal end of the handle **70**. This first injection adapter **99a** is used for flushing the insertion system **100** and as outlet of a guide wire **180**, with the aid of which the actual introduction of the catheter shaft **30** with the catheter tip **10** provided at the distal end of the catheter shaft **30** into the body of the patient is simplified. The catheter shaft **30**, the catheter tip **10** and the handle **70** are thereby threaded into the guide wire **180** and pushed along it, for example into the aorta and to the heart of the patient.

(133) In the embodiments of the insertion system **100** designed for transarterial/transfemoral access, a second injection adapter **99b** may further be provided, by means of which a liquid coolant

etc. can be passed, for example, via the fluid channels **43** (see FIG. **10a**) formed in the interior of the catheter shaft **30** to the catheter tip **10**. With the aid of such a liquid coolant, a stent accommodated in the catheter tip **10** can be appropriately cooled while the catheter tip **10** is being advanced to the implantation location, as long as the insertion system **100** is in its first functional state, in which the catheter tip **10** is completely enclosed by the telescopically arranged sleeve-shaped members **11** and **21**.

(134) The provision of cooling that can be produced with the second injection adapter **99b** for the stent accommodated in the catheter tip **10** is a particular advantage when a shape memory material is used as stent material and when the stent can deform under the effect of an external stimulus from a temporary form to a permanent form, whereby the temporary form exists in the first configuration of the stent (in the folded-up state, when the stent is accommodated in the catheter tip **10**) and the permanent form exists in the second configuration of the stent (in the expanded state of the stent after release of the stent from the catheter tip **10**).

(135) In the embodiments of the insertion system **100** previously described, the guiding tube **52** is preferably made from a material allowing the guiding tube **52** to be capable of traversing a tortuous pathway in the body of the patient without kinking. For example, the guiding tube **52** may include an inner lubricious liner, an outer polymeric jacket, and a coil reinforcement between the inner and outer layers. In addition, it is preferred when at least one radiopaque band or member is incorporated within the guiding tube's material to allow precise location of the distal end of the guiding tube **52** for positioning accuracy.

(136) On the other hand, the first and second catheter tubes **32**, **42** of the catheter shaft **30** are preferably made from flexible, sterilizable materials. These materials may include, for example, polyurethane, silicone, polyvinyl chloride (PVC) nylon and/or polyether block amide, e.g. Pebax®. Furthermore, the first catheter tube **32** and/or second catheter tube **42** are/is at least partly made from a less rigid material than the guiding tube **52**. In an exemplary embodiment, the first catheter tube **32** and/or the second catheter tube **42** are/is at least partly made of a braided wire construction. In addition, the stent holder tube **62** may also be at least partly made of a braided wire construction.

(137) Individual features of different embodiments of this disclosure may be combined in any suitable manner.

(138) A preferred embodiment of a medical device for treatment of a heart valve stenosis and/or heart valve insufficiency in a patient is described in the following with reference to FIGS. **12a** to **12c**. As depicted, the medical device exhibits an insertion system **100** designed for transarterial transfemoral access, as has been described in detail previously, for example, with reference to FIGS. **1** to **10**.

(139) In addition to the insertion system **100**, the medical device has an expandable heart valve stent **150** mounted in the catheter tip **10** of the insertion system **100**, to which a heart valve prosthesis **160** to be implanted is fastened. In the first functional state, not shown, the stent **150** exhibits a first, previously definable configuration, in which it is in its folded-together state. On the other hand, the stent **150** is designed to adopt a second previously definable configuration in the implanted state, in which it exists in its expanded state.

(140) Through the use of the insertion system **100** described above, during the implantation procedure, the stent **150** is transferred sequentially, following a previously definable sequence of events in steps from its first previously defined configuration into its second previously defined configuration.

(141) In detail, the stent **150** that is used with the medical device in accordance with the depiction in FIGS. **12a** to **12c** exhibits a first retaining region, to which the heart valve prosthesis **160** is attached. Further, the stent **150** comprises a second retaining region with three retaining elements **151**, each in the configuration of retaining rings, which can be brought in to a releasable engagement with the retaining elements **16** of the stent holder **15** provided in the catheter tip **10**.

(142) In addition, the stent **150** has three retaining hoops **153** to accommodate the heart valve

prosthesis **160** and three positioning hoops **154** for automatic positioning of the stent **150** at the implantation site, whereby the respective positioning hoops **154** of the stent **150** are designed in functional and structural respects to engage the pockets **170** of the native heart valve during the implantation procedure and in the implanted state of the stent **150**, in particular from the second functional state of the insertion system **100**. In detail, each positioning hoop **154** and its associated retaining hoop **153** has an essentially U or V-shaped structure, which is closed towards the distal end of the stent **150**.

(143) The stent **150**, which together with the insertion system **100** forms the basis of the medical device, is especially suitable for insertion into the body of a patient with the aid of the insertion system **100** with minimal invasiveness. The distinctive feature of the stent **150** is that the three positioning hoops **154** of the stent **150** undertake the function of automatic positioning of the stent **150** with the heart valve prosthesis **160** attached to it in the aorta of the patient. The positioning hoops **154** have radiused head sections, which engage in the pockets **170** of the insufficient heart valve to be replaced by the heart valve prosthesis during positioning of the stent **150** at the implantation site. The provision of a total of three positioning hoops **154** takes care of the necessary positioning accuracy in the rotary direction.

(144) In this state shown in **12a**, the catheter tip **10** and the catheter shaft **30** of the transarterial or transfemoral insertion system **100** has been inserted by a puncture of the groin artery of the patient and the catheter tip **10** has been advanced to the implantation site with the aid of a guide wire **180**. In detail, the insertion system **100** to be used is shown already in its second functional state in FIG. **12a**. The second functional state of the insertion system **100** designed for transarterial or transfemoral access has been described previously, for example with reference to FIG. **7b**.

(145) In the second functional state, the first sleeve-shaped member **11** of the catheter tip **10** has already moved by a first predetermined amount of movement in a proximal direction, and thus towards the handle **70**, leading to a release of the positioning hoops **154** of the stent **150**. These already expanded positioning hoops **154** of the stent **150** shown in FIG. **12a** are positioned—where necessary by a suitable rotation of the stent holder **15** of the catheter tip **10**—in the pockets **170** of the native heart valve position. After positioning of the positioning hoops **154** in the pockets **170** of the native heart valve is complete, the insertion system **100** is transferred from its second functional state (see FIG. **7b**) into its third functional state (see FIG. **7c**).

(146) The manner in which the insertion system **100** is transferred into its third functional state has been described previously, for example with reference to FIG. **7c**, FIG. **12b** shows the insertion system **100** in accordance with FIG. **12a**, in which the second sleeve-shaped member **21** has been displaced in a distal direction so that the first retaining region of the stent **150** with the retaining hoops **153** and the heart valve prosthesis **160** attached to them are released. These components are opened as a result of the radial forces attacking them, whereby the old heart valves are clamped between the positioning hoops **154** and the retaining hoops **153**.

(147) After the functioning of the heart valve prosthesis **160** has been checked, the insertion system **100** is then transferred from its third functional state into its fourth functional state, as has previously been described, for example with reference to FIG. **7d**. FIG. **12** shows the effect of the transfer of the insertion system **100** into its fourth functional state on the heart valve prosthesis **160** and the stent **150**.

(148) In detail, it can be seen that, in the fourth functional state of the insertion system **100**, the first sleeve-shaped member **11** of the catheter tip **10** has been displaced further in a proximal direction, as a result of which the anchorage of the retaining elements **151** on the second retaining region of the stent **150** is released. This has the result that the second retaining region of the stent **150** can also expand and press against the vessel wall.

(149) Finally, the catheter tip **10** and the catheter shaft **30** of the insertion system **100** are removed again from the body of the patient.

(150) When the heart valve stent **150** is implanted, the old (insufficient) heart valve is pressed

against the vessel wall at the same time due to the self-expanding characteristic of the stent **150**, as can be seen in particular in FIG. **12c**. In particular, the semilunar heart valves of the insufficient, native heart valve are damped between the positioning hoops **154** and the retaining hoops **153** because of the expansion of the stent **150**, in addition to which the heart valve prosthesis **160** located on the first retaining region of the stent **150** is optimally positioned and is stably anchored. (151) The disclosed solutions provide an improved insertion system **100** with the stent mountable in the catheter tip **10** of the insertion system **100**. The stent may be inserted transarterially by the special insertion system **100** and can be optimally positioned, so that a heart valve prosthesis sewn on the first retaining region of the stent can undertake the function of the insufficient or stenosed native heart valve. The radial forces developed due to the self-expanding characteristic of the stent ensure a secure anchoring in the area of the aorta. The catheter system **1** of the insertion system **100** is preferably an 18 to 21F introducer, which is compatible with 21F-insertion tubes and a 0.035" guide wire **180**. The length of the catheter system **1** for transarterial access should be at least 100 cm. The optionally provided flexural link region at the distal region of the catheter system **1** is preferably approximately 30 cm.

(152) A further embodiment of a catheter tip **10** for an insertion system for transfemoral/transarterial insertion of an expandable heart valve stent is shown in its four different functional states in FIGS. **13a** to **13d**. In detail, the catheter tip **10** is shown in its first functional state in FIG. **13a**, in which the catheter shaft with the catheter tip **10** and, where required, with the stent accommodated in it can be inserted into the patient transarterially or transfemorally and advanced via the aorta to the implantation site.

(153) In the first functional state of the catheter tip **10** in accordance with FIG. **13a**, the catheter tip **10** is completely closed, whereby the two sleeve-shaped members **11**, **21** of the catheter tip **10** abut. In this embodiment, the two sleeve-shaped members **11**, **21** of the catheter tip **10** have an equal outer cross-section diameter, thereby not forming a step in the state depicted in FIG. **13a**. The respective inner diameters of the sleeve-shaped members **11**, **21** are chosen so that the folded-up retaining hoops of a stent, with the heart valve prosthesis fastened to them where required, can be housed in the second sleeve-shaped member **21**. The folded-up positioning hoops of the stent housed between the second sleeve-shaped member **21** and the first sleeve-shaped member **11** are held together in their folded form.

(154) In the first functional state of the catheter tip **10**, as shown in FIG. **13a**, the second retaining region of the stent is fixed with the stent holder **15** at the proximal end of the catheter tip **10**. For this purpose, the retaining elements (retaining rings etc.) provided at the second retaining region of the stent are engaged with retaining elements **16** of the stent holder **15**.

(155) The retaining elements **16** of the stent holder **15** are covered by the first sleeve-shaped member **11** of the catheter tip **10** in the first functional state shown in FIG. **13e**, so that an engagement between retaining elements provided on the second retaining region of a stent and retaining elements **16** of the stent holder **15** would be possible.

(156) The first functional state of the catheter tip **10** shown in FIG. **13e** is maintained during the transarterial insertion procedure. On reaching the implantation location, the catheter tip **10** is transferred from the first functional state shown in FIG. **13a** to the second functional state shown in FIG. **13b**, by transferring the first operating means of the handle (first operating means **71** shown in the embodiment of the wheel in FIG. **7**) from the first position into the second position. The longitudinal displacement stroke transferred by actuation of the first operating means **71** to the first sleeve-shaped member **11** of the catheter tip **10** effects a displacement of the first sleeve-shaped member **11** relative to the stent holder **15** in the proximal direction, thus towards the handle **70**.

(157) The longitudinal displacement stroke executed on the first sleeve-shaped member **11** of the catheter tip **10** during the transition from the first functional state (see FIG. **13a**) to the second functional state (see FIG. **13b**) by the first operating means **71** of the handle **70** via a corresponding first force transmission means **31** is previously defined so that the first sleeve-shaped member **11** is

displaced relative to the stent holder **15** in the proximal direction just so far that the positioning hoops of a stent housed in the catheter tip **10** would be released, though the distal end of the first sleeve-shaped member **11** of the catheter tip **10** would still cover the retaining elements **16** of the stent holder **15**, so that the engagement between the retaining elements provided at the second retaining region of the stent and the retaining elements **16** of the stent holder **15** would be secure. (158) Since the second sleeve-shaped member **21** is not manipulated during the transition from the first functional state into the second functional state, the first retaining region of a stent housed in the catheter tip **10** with the heart valve prosthesis fastened to it would continue to be housed in its folded together state in the sleeve-shaped element of the second sleeve-shaped member **21**.

(159) The positioning hoops of a stent housed in the catheter tip **10** released in the second functional state of the catheter tip **10** are opened as a result of the radial forces acting on them and can thus be positioned in the pockets of the insufficient native heart valve. Following appropriate positioning of the positioning hoops of the stent in the pockets of the native heart valve, the catheter tip **10** is transferred from the second functional state shown in FIG. **13b** into the third functional state shown in FIG. **13c**. This is done by manipulation of the second operating means **81** of the handle, after the securing element **89** associated with the second operating means **81** has been removed.

(160) On actuation of the second operating means **81** of the handle, the second sleeve-shaped member **21** of the catheter tip **10** associated with the second operating means **81** is moved relative to the stent holder **15** by a previously established longitudinal displacement stroke defined with the second operating means **81** in a distal direction, thus away from the handle **70**. The longitudinal displacement stroke acting on the second sleeve-shaped member **21** is chosen so that the sleeve-shaped member **21** no longer covers the first retaining region of a stent housed in the catheter tip **10** with the heart valve prosthesis fastened to it, where required, and thus releases the first retaining region of the stent. Due to the action of the radial forces, the distal retaining region of the stent with the heart valve prosthesis attached to it, where required, unfolds completely.

(161) Since the first operating means **71** of the handle **70** and the associated first sleeve-shaped member **11** of the catheter tip **10** are not manipulated during the transition from the second functional state in accordance with FIG. **13b** into the third functional state in accordance with FIG. **13c**, the distal end region of the first sleeve-shaped member **11** continues to cover the retaining elements **16** of the stent holder **15**, so that the engagement between the retaining elements of a stent housed in the catheter tip **10** and the retaining elements **16** of the stent holder **15** is secure and the proximal retaining region of the stent is in its folded-up state. This anchorage of the stent to the catheter tip **10** of the insertion system **100** allows an explantation of a stent that is already partially unfolded by returning the catheter tip **10** from the third functional state, by appropriate manipulation of the second operating means **81** of the handle **70**, to the second functional state and then by suitable actuation of the first operating means **71** transfer to the first functional state.

(162) If an explantation of the stent with the heart valve prosthesis attached to it, where required, is unnecessary, the catheter tip **10** is transferred from the third functional state shown in FIG. **13c** into the fourth functional state shown in FIG. **13d**, by turning the first operating means **71** of the handle **70** further from the second position to the third position after removal of the securing element **79** (locking element). This manipulation of the first operating means **71** that can be effected after removal of the securing element **79** results in a further defined movement of the first sleeve-shaped member **11** relative to the stent holder **15** of the catheter tip **10** in a proximal direction, thus towards the handle **70**. The longitudinal displacement stroke executed on the first sleeve-shaped member **11** is chosen so that the distal end of the first sleeve-shaped member **11** no longer covers the retaining elements **16** of the stent holder **15**, as a result of which an engagement between the retaining elements of a stent housed in the catheter tip **10** and the retaining elements **16** of the stent holder **15** can be released, which would also lead to a complete release of the second retaining region of the stent and a complete separation of the stent from the catheter tip **10** and correspondingly to a

complete unfolding of the stent.

(163) In the embodiment of the catheter tip **10** depicted in FIGS. **13a-e**, a stent holder tube **62** is used for connecting the stent holder **15** of the catheter tip **10** to the body **70'** of the handle **70**. The stent holder tube **62** has a distal end connected to the stent holder **15**, a proximal end connected to the body **70'** of the handle **70** and a passageway extending through the stent holder tube **62**. In addition, an extension portion **62'** of the stent holder tube **62** is provided, said extension portion extending from the distal end of the stent holder **15** to a support section **63**. The support section **63** may be a tapered portion which is completely accommodated in the second sleeve-shaped member **21** when the catheter tip **10** is in its first and second functional state (cf. FIGS. **13a, b**).

(164) Preferably, the stent holder tube **62** and its extension **62'** have a cross-section less than the cross-section of the first catheter tube **32** and greater than the cross-section of the second catheter tube **42** (not shown in FIGS. **13a-e**), wherein the first catheter tube **32** is disposed concentrically and coaxially with the stent holder tube **62** thereby accommodating the stent holder tube **62** such that the first catheter tube **32** is moveable relative to the stent holder tube **62**. The passageway of the stent holder tube **62** shall have a diameter sufficient to accommodate the second catheter tube **42** such that the second catheter tube **42** is moveable relative to the stent holder tube **62**.

(165) FIG. **13e** shows a side elevation of the embodiment of the catheter tip **10** in accordance with FIG. **13a-d**, whereby the catheter tip **10** is in its state after releasing a stent housed in the catheter tip **10** and ready to be removed again from the body of the patient. In this state of the catheter tip **10**, the first sleeve-shaped member **11** is pushed by manipulation of the first operating means **71** of the handle **70** such that the first sleeve-shaped member **11** is in its most distal position, in which the distal end of the first sleeve-shaped member **11** abuts against the proximal end of the second sleeve-shaped member **21** without any gap or step there between. For securing this gap and step free state, the distal end of the first sleeve-shaped members **11** is supported by the already mentioned support section **63**.

(166) The disclosed solution is not limited to the preferred embodiment described in the attached drawings. On the contrary, combinations of the individual features described in detail are also possible.

(167) TABLE-US-00001 List of reference numerals 1 catheter system 10 catheter tip 11 first sleeve-shaped member 15 stent holder 16 retaining elements 21 second sleeve-shaped member 25 catheter end tip 30 catheter shaft 31 first force transmission means 32 first catheter tube 34 flexural link region 36 channel 35 control wire 40 filler body 41 second force transmission means 42 second catheter tube 43 fluid channels 51 guiding means 52 guiding tube 53 inlet port 62 stent holder tube 62' extension of stent holder tube 63 support means 70 handle 70' body of the handle 71 first operating means 72 first guide 73 first pusher 74 first slide 74' screw cap 75 first stop 75 second stop 77 additional stop 77' locking element 79 securing element 81 second operating means 82 second guide 83 second pusher 84 second slide 85 first stop 86 second stop 87 fixing 88 grip 89 securing element 96 third operating means 97 compression spring 97a first stop 97b second stop 98 turning mechanism/rotatable cap 99a first injection adapter 99b second injection adapter 100 insertion system 150 stent 151 retaining elements 153 retaining hoops 154 positioning hoops 160 heart valve prosthesis 170 pockets of native heart valve 180 guiding wire L longitudinal direction of insertion system 100

Claims

1. A delivery system for introducing an expandable stent having a distal end and a proximal end into a heart of a patient, the delivery system comprising a distal end portion comprising: a catheter tube; an end support coupled to a distal end of the catheter tube, a proximal surface of the end support being tapered; a stent holder comprising an outer surface that is tapered, the stent holder positioned at the distal end portion and configured to directly mate with a proximal end of the

expandable stent such that a distal end of the expandable stent is disposed closer to the end support than the proximal end of the expandable stent; and a sheath movable relative to each of the catheter tube and the stent holder; a handle at a proximal end of the delivery system, the handle configured to manipulate at least the stent holder, wherein the distal end portion of the delivery system is configured to house the expandable stent comprising at least two positioning arches in a collapsed configuration such that the expandable stent is releasably engaged with the stent holder at the proximal end of the expandable stent, wherein the handle comprises at least one actuator for sequentially releasing the expandable stent from the distal end portion, and wherein the at least one actuator is configured to permit the at least two positioning arches to expand while at least a portion of the expandable stent remains in the collapsed configuration.

2. The delivery system of claim 1, wherein the distal end portion further comprises an end tip that is cone-shaped.

3. The delivery system of claim 1, wherein the catheter tube defines a lumen configured to receive a guidewire.

4. The delivery system of claim 1, further comprising a shaft positioned between the handle and the distal end portion, wherein the shaft is flexible thereby permitting the distal end portion to navigate the aortic arch.

5. The delivery system of claim 1, wherein the handle is configured to rotate the distal end portion about a longitudinal axis of the distal end portion.

6. The delivery system of claim 1, wherein the stent holder is configured to releasably engage retaining elements extending from a proximal end of the expandable stent.

7. The delivery system of claim 1, wherein the distal end portion comprises at least one radiopaque marker for positioning the distal end portion in the aortic valve.

8. A method for introducing an expandable stent having a distal end and a proximal end into a heart of a patient, the method comprising: advancing a distal end portion of a delivery system through an aortic arch of a patient towards an aortic valve of the patient, the distal end portion comprising: a catheter tube; an end support coupled to a distal end of the catheter tube and comprising a proximal surface that is tapered; a stent holder comprising an outer surface that is tapered, the stent holder positioned at the distal end portion and configured to directly mate with a proximal end of the expandable stent such that a distal end of the expandable stent is disposed closer to the end support than the proximal end of the expandable stent; and a sheath movable relative to each of the catheter tube and the stent holder; a handle at a proximal end of the delivery system, the handle configured to manipulate at least the stent holder and comprising at least one actuator for sequentially releasing the expandable stent from the distal end portion; introducing the distal end portion through the aortic valve, wherein the expandable stent comprises at least two positioning arches and is housed in the distal end portion in a collapsed configuration such that the expandable stent is releasably engaged with the stent holder at the proximal end of the expandable stent; manipulating the at least one actuator of the handle to permit the at least two positioning arches to expand while at least a portion of the expandable stent remains in the collapsed configuration; and releasing the expandable stent from the stent holder to cause the expandable stent to transition to an expanded configuration within the aortic valve.

9. The method of claim 8, wherein releasing the expandable stent comprises manipulating the stent holder using the handle to release the expandable stent.

10. The method of claim 8, wherein the delivery system further comprising a flexible shaft positioned between the handle and the distal end portion, and further comprising using the handle to cause the flexible shaft to bend to navigate the aortic arch.

11. The method of claim 8, further comprising sequentially releasing the expandable stent from the distal end portion using the handle.

12. The method of claim 8, further comprising rotating the distal end portion about a longitudinal axis of the distal end portion using the handle.

13. The method of claim 12, further comprising aligning the expandable stent with the aortic valve.
 14. The method of claim 8, wherein releasing the expandable stent from the stent holder comprises releasing retaining elements extending from a proximal end of the expandable stent.
 15. The method of claim 8, wherein the distal end portion further comprises at least one radiopaque marker, and further comprising visualizing the distal end portion in the aortic valve using the at least one radiopaque marker.
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